

THE ENERGY BALANCE PROJECT

Mathematical Structure and Proof

Part II

Revised Gaussian Explanatory Edition

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About this revised edition

The first volume asks what a physical account should describe. This volume derives the mathematical statements that such an account can honestly make. Its active worked model is the Local Gaussian specialization: a fixed quadratic deviation potential on a conserved, nonnegative scalar resource. The potential supplies a valuation, not an automatic action selector.

This edition revises the original unified explanatory Part II. It preserves its patient approach to notation, derivations, assumptions and worked calculations. The original editable manuscript was not recovered; the active chapters here are reconstructed editable source, not a facsimile. The supplied original PDF remains unchanged. Selected original proofs appear at the end as a clearly labelled historical appendix.

Chapter numbering follows the revised Book I: this volume begins at 33 and ends at 46. Original chapter and equation numbers survive only inside the historical appendix, where they identify the old model. A repository disposition ledger maps every original outline entry to its revised subject or preserved historical home. Do not confuse an old equation number with a new one simply because both books use the word EBU.

Four different kinds of statement

A definition declares a model. A proof establishes a consequence of explicit assumptions. An implementation carries out a specified operation. An experiment supplies evidence about a specified system. None becomes another merely by being written in mathematical notation or committed to a repository.

The Gaussian programme remains prospective. Its capacity rule, ownership map, event profile and random selection semantics require their own acceptance. The conditional derivations below do not adopt them by implication. This book introduces no scientific run, no parameter calibration and no claim of economic or ecological success. It is not independently peer reviewed.

How to read the proofs

Read Chapters 33–35 if vectors, units or action graphs are unfamiliar. They turn a physical story into declared objects without importing a controller. Chapters 36–40 derive the Gaussian field, exact finite accounting, locality, path integration and telescoping. Chapters 41–43 address groups, attribution and the proposed capacity account. Chapters 44–45 explain why a correct account is not yet a theorem about dynamics. Chapter 46 combines the ideas in a worked laboratory, with deliberately adverse as well as favorable cases.

A displayed equality is worth reading in two passes. First identify what every term denotes and which state it uses. Then check the algebra. Most serious accounting mistakes in this subject are not failures to multiply: they compare different baselines, mix rates and quantities, omit an affected factor, or add an institutional meaning that the equality did not prove.

The examples use small, exact numbers so the reader can reconstruct them. They are illustrative arithmetic, not sampled trajectories. When a later chapter writes a candidate transition rule, that is a specification on paper; it is not a report that the research engine was executed to produce it.

The historical appendix preserves the original continuous-time, timestep, network-certificate and P1C derivations. Read its chapter numbers as archival identifiers. Its flow law is not the random-affordable actor process described in the active programme. The new Chapter 44 explains the bridge and its limits.

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From a physical story to a mathematical claim

A store has more water than its declared reference and a clinic has less. Moving water from one to the other seems helpful. Mathematics begins by asking what that sentence has left unspecified. How much water? Which reference? Which route? What else changes? Does the action remain helpful when it is large? Who is permitted to perform it? The value of a formal model is not that it makes these questions disappear. It gives each question an object precise enough to inspect.

33.1 The declared world

Let there be a finite number n of accounting cells, with state

$$x = (x_1, \dots, x_n)^T \in \mathcal{X}_M, \quad \mathcal{X}_M = \{x \in \mathbb{R}^n : x_i \geq 0, \sum_i x_i = M\}. \quad (33.1)$$

Every x_i measures the same homogeneous resource. The superscript T turns a written row into a column. The set $\mathcal{X}_M$ is the nonnegative fixed-mass simplex: all ways of distributing the same total M among the cells. For two cells it is a line segment, not the whole plane. For three it is a triangle embedded in three-dimensional space.

The programme declares a reference $x^* \in \mathcal{X}_M$ and scales $\sigma_i > 0$. Membership of the reference in the same simplex matters. If two cells contain twenty units in total, references requiring twelve units in each cannot be reached by closed internal transfer. That mismatch is not an actor failure. It is a different modelling problem.

Compatibility is necessary, not sufficient, for reachability. A graph with no route to the clinic cannot deliver water there. A menu containing only four-unit transfers may be unable to remove a one-unit discrepancy exactly. The geometry says which states are allowed by total mass; the action system says which transitions are available.

33.2 Definition, proposition and evidence

We define a potential and derive its consequences. A definition is a choice about the model, not a discovery that nature uses that formula. A proposition may then be true for every state in the declared domain. A worked example can help a reader follow the proof but cannot replace it. An experiment would examine a specified transition process under registered conditions.

The programme document [3] makes this distinction explicit. The active model is prospective. Its exact quadratic identities can be proved now. Whether a random-affordable actor process produces useful recovery, distribution or long-run behavior is a different question.

Standing assumptions for active examples

One homogeneous conserved scalar; a finite number of cells; nonnegative stocks; fixed compatible references; fixed positive scales; lossless internal actions; additive declared increments. Unless a section explicitly says otherwise, separate process burden is zero. No restoring dynamics or optimizing actor is hidden in these assumptions.

33.3 What the state leaves outside

A simple water stock model does not automatically represent water quality, pumping energy, soil damage or a patient's health. These omissions limit the claim. They cannot be repaired by saying the scalar potential is holistic. If quality is consequential, it needs a represented coordinate or an explicitly scoped limitation. If pumping consumes electricity, a broader physical account must link that transformation to an energy carrier.

Nor does the model identify a reference with a moral optimum. A reference may later be informed by functional evidence and public choices, but the proof that follows only knows its declared number. Keeping this boundary visible lets mathematical work proceed without pretending that calibration or governance has already been solved.

33.4 A useful proof-reading habit

Before accepting a theorem, write its domain in words. When we later prove that V is bounded, the reason will be finite mass and fixed scales, not an EBU controller. When we prove that a sequence telescopes, the reason will be consecutive endpoint cancellation, not benevolent actors. When we prove

that path contributions sum to the group value, the reason will be linearity and the fundamental theorem of calculus, not a fair ownership rule.

This habit prevents a correct sentence from acquiring an incorrect sequel. “The accounts close” does not imply “the world recovers.” “The action has positive value” does not imply “the action is physically possible.” Each sequel needs its own premises.

Exercises and discussion

For a three-cell world with mass twelve, decide whether $(4, 4, 4)$, $(5, 5, 5)$ and $(12, 0, 0)$ are compatible references. The first and third are; the second is not. Compatibility of the third still says nothing about whether the graph can move all stock to cell one.

Explain why the phrase “Gaussian equilibrium” is potentially misleading. The Gaussian construction defines reference geometry. To call the reference an equilibrium of a process, one must specify that process and show that it leaves the reference unchanged. To call it attracting requires more still.

Chapter recap

A model needs a domain, a state, a reference, a transformation and a claim. The rest of this volume makes those objects explicit. Its mathematical results are conditional statements about the declared objects, not a shortcut around physical measurement or scientific testing.

Units, vectors and conservation

Many apparent disagreements about EBU are disagreements about units. One author writes q for a rate, another for a finite quantity; both use the same symbol in a correct-looking equation. The error appears only when the duration differs from one. We will therefore identify quantity before performing algebra.

34.1 A dimensional dictionary

Write X for the unit of physical stock and T for time. In the active Gaussian normalization, V is dimensionless, as are its finite differences. Calling that numerical accounting unit EBU does not make it a joule, a litre or a currency unit. A later calibrated conversion must be declared rather than inferred from the name.

Object	Unit and meaning
x_i, x_i^*, σ_i, q	X : stock, reference, scale, finite withdrawal
j	X/T : a rate
$\mu_i = \partial V / \partial x_i$	$1/X$: potential change per unit stock
$H_{ii} = 1/\sigma_i^2$	$1/X^2$: curvature
$q\mu_i, q^2 H_{ii}$	dimensionless finite-account terms

If a device operates at constant rate j for duration Δt , its quantity is $q = \Delta t j$. Thus $\delta = Sq$ and $\delta = \Delta t S j$ are equivalent descriptions, not two steps to apply successively. The historical foundation uses rate notation [4]; this edition uses finite quantities in its active examples.

34.2 Reading a matrix multiplication

For a lossless chain $1 \rightarrow 2 \rightarrow 3$, let

$$S = \begin{pmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad q = \begin{pmatrix} a \\ b \end{pmatrix}. \quad (34.1)$$

Then

$$Sq = \begin{pmatrix} -a \\ a - b \\ b \end{pmatrix}. \quad (34.2)$$

The first cell loses a , the middle receives a and sends b , and the last receives b . The middle coordinate is not a transport label that may be omitted: it is a physical stock with its own predecessor and successor.

Every column of S sums to zero. If $\mathbf{1}$ is the vector of ones, $\mathbf{1}^T S = 0$, so

$$\mathbf{1}^T(x + Sq) = \mathbf{1}^T x. \quad (34.3)$$

This proves conservation for the declared lossless increments. It does not prove nonnegativity: a request $a > x_1$ still conserves the algebraic total while making the source negative. Conservation and admissibility must be checked separately.

34.3 Loss is a physical account, not a vanished number

For a lossy transfer, a column may have -1 at the source and η at the destination. Its sum is $-(1 - \eta)$. To close a broader physical account, the missing quantity must enter a declared loss carrier or cross an explicit boundary. For instance an added coordinate ℓ can receive $(1 - \eta)q$. That is a conservation declaration, not yet a valuation of ℓ or permission to charge a second process cost.

The active first domain sets $\eta = 1$. This does not dismiss loss. It avoids claiming that an unresolved loss-aware runtime and valuation convention has already been accepted. Part III returns to the old Definition 6.4 wording: adding a loss coordinate changes what can honestly be said about a carrier being absent from state.

34.4 Changing the physical unit

Suppose litres are replaced by millilitres. Multiply all stocks, references, scales and finite quantities by $c = 1000$. Each standardized deviation remains unchanged:

$$\frac{cx_i - cx_i^*}{c\sigma_i} = \frac{x_i - x_i^*}{\sigma_i}. \quad (34.4)$$

Consequently V and exact EBU are unchanged. The marginal becomes μ_i/c , while q becomes cq ; their product is unchanged. This is a useful dimensional check on an implementation specification.

Changing σ while keeping stocks fixed is not a unit conversion. It changes the model's relative sensitivity. Treating those operations as the same would hide a substantive calibration decision inside a formatting step.

Worked checks

A rate of six units per hour over a quarter hour transfers $3/2$ units. Over two hours it transfers twelve. A routine that uses six in both cases has silently assumed a unit-duration interval.

In the chain matrix, take $a = 3$, $b = 2$ and predecessor $(5, 1, 0)$. The declared endpoint is $(2, 2, 2)$, with total six before and after. Whether the middle cell may send two while initially holding one depends on timing: one may allow a declared inflow-funded simultaneous process, or require outgoing commitments to be covered by frozen stock. The matrix alone cannot settle that physical convention.

Chapter recap

Vectors collect quantities; matrices express declared transformations. Units and column sums expose important errors, but they do not certify the whole physical process. Conservation, nonnegativity, timing and valuation each need their own check.

Graphs, supports and physically admissible groups

An edge is a possible physical transformation, not a promise that a helpful transfer can always occur. A route, a pump and a warehouse doorway may each impose limits independent of any potential. Begin with those limits, because an exact value for an impossible action is only a counterfactual calculation.

35.1 Three graphs with different jobs

The physical graph lists allowed transformations. The potential-factor graph lists which state coordinates a term in V depends on. The action-interaction graph records overlap or coupling between actions. In the diagonal Gaussian model every potential factor has one coordinate, yet two physical actions can interact because they both change the same source. A diagonal Hessian does not make all edge actions independent.

For action a , its write support is the set of coordinates with nonzero increment δ_a . Its constraint support includes capacities, reservations or aggregate budgets it uses. Two actions may have disjoint write supports but share a machine capacity. Conversely, reading the same immutable scale does not by itself create a shared physical constraint.

35.2 Joint feasibility is not singleton feasibility

Suppose Ridge contains five units. Two requests each withdraw three units from Ridge to different destinations. Each is individually nonnegative at its endpoint, but their group withdraws six and is invalid. Testing children one at a time against the same five units does not validate their sum.

With $q_a \geq 0$, a conservative frozen-stock rule might require

$$\sum_{a: \text{src}(a)=i} q_a \leq z_i. \quad (35.1)$$

This is an example of a declared rule, not a newly adopted programme policy. A plant that permits concurrent incoming flow to fund outgoing flow requires

a different timing and feasibility contract. Endpoint nonnegativity alone does not prove that an unmodelled intermediate mechanism can perform it.

Destination capacity is another independent constraint. If a tank can hold ten and begins at eight, two incoming deliveries of two each cannot both fit merely because their separate endpoints do. Physical capacity is not the Gaussian reference. A reference may be six even when the tank can hold ten; exceeding the reference can be feasible but increase potential.

35.3 The straight segment and its limited guarantee

The simplex is convex. If z and $z + \delta_G$ lie in it, then

$$z + s\delta_G = (1 - s)z + s(z + \delta_G), \quad 0 \leq s \leq 1, \quad (35.2)$$

also lies in it. This proves nonnegative conserved stock along the aggregate straight interpolation. It does not prove child-specific rate limits, mechanical synchronization, queue clearance or causal access to incoming stock. Those are additional constraints on a physical realization.

This distinction matters when a path integral is later described as physical. The same segment can be a valid mathematical interpolation even when the plant has not declared constant-rate joint execution. The integral identity survives; the physical-path interpretation does not follow automatically.

35.4 Grouping without losing atomic actions

A joint group retains child identities and increments. It is not an excuse to conceal a fuel-to-electricity-to-pump-to-water chain inside one undifferentiated operation. Distinct carriers and intermediate transformations must remain traceable when they belong to the represented scope.

The analytical bridge [5] groups coupled actions under a common accounting boundary and distinguishes physical grouping from receipt batching. Batching independent records can reduce administrative overhead, but it does not create a new physical synergy. Conversely, simultaneous actions sharing a source need joint evaluation even if their paperwork is submitted separately.

Exercises with reasoning

Take a source with six units and two proposed quantities (4, 4). A common cap of six with proportional scaling would yield (3, 3). The number three comes from that resolver, not from the value equation. If the resolver were priority-based, a different accepted vector could be physically admissible. Never price (4, 4) and execute (3, 3) under the old value record.

Now let two accepted actions use disjoint nodes and disjoint constraints. With a separable potential, their finite values add because the affected terms are disjoint. If a new pair factor couples one node from each action, the independence proof no longer applies even though the physical edges remain disjoint. The correct affected support follows the potential factors.

Chapter recap

The graph says what can connect, support says what changes, constraints say what is permitted, and the potential says how the represented change is valued. None can silently perform the job of another.

Gaussian reference geometry

The word Gaussian can suggest random noise, a bell-shaped histogram or a statistical fit. Here its primary role is geometric. We declare a reference and a positive scale at each cell, measure deviation in units of that scale, square it and add. No observations have to be normally distributed for that mathematical definition to make sense.

36.1 The potential and its matrix form

Define

$$V(x) = \frac{1}{2} \sum_{i=1}^n \frac{(x_i - x_i^*)^2}{\sigma_i^2} = \frac{1}{2} (x - x^*)^T H (x - x^*), \quad H = \text{diag}(\sigma_i^{-2}). \quad (36.1)$$

The factor $1/2$ simplifies derivatives. Each diagonal entry of H is positive. Thus for a nonzero vector v , $v^T H v = \sum_i v_i^2 / \sigma_i^2 > 0$. This is positive definiteness, and it gives $V \geq 0$ with equality exactly at $x = x^*$.

Because the reference is assumed compatible with the physical domain, it is also the unique minimizer on that domain. A minimum of a function is not yet an attracting state of a process. It is a geometric fact available before an actor law has been specified.

36.2 Deriving the marginal and curvature

Differentiate one summand. The derivative of $(x_i - x_i^*)^2$ is $2(x_i - x_i^*)$, cancelling the factor $1/2$. Hence

$$\mu_i(x_i) = \frac{x_i - x_i^*}{\sigma_i^2}, \quad \mu(x) = H(x - x^*), \quad \nabla^2 V = H. \quad (36.2)$$

The marginal is negative below reference and positive above it. There is no flat healthy band. A stock very close to its reference has a small, usually nonzero marginal. This is a substantive departure from the old hinge family, not a cosmetic rewriting of its coefficients.

Let $h_{\min} = \min_i \sigma_i^{-2}$ and $h_{\max} = \max_i \sigma_i^{-2}$. Then

$$\frac{h_{\min}}{2} \|x - x^*\|_2^2 \leq V(x) \leq \frac{h_{\max}}{2} \|x - x^*\|_2^2. \quad (36.3)$$

These inequalities follow by replacing each positive diagonal weight with its smallest or largest value. They relate potential to Euclidean distance under this declared calibration. They do not say that Euclidean distance is a complete measure of human or ecological function.

36.3 What changing a scale does

Take $(x_1, x_2) = (18, 2)$, reference $(10, 10)$ and scales $(2, 2)$. The standardized deviations are $(4, -4)$ and $V = 16$. If only the first scale becomes one, the potential becomes $\frac{1}{2}(8^2 + (-4)^2) = 40$. The physical state has not changed; the declared sensitivity has. That is why comparisons across parameter versions need explicit recalibration records.

The shape $\exp[-V(x)]$ resembles a Gaussian density, but it is not itself a normalized probability law. Restricting to fixed mass couples the coordinates, and the nonnegative boundary further changes the domain. A claim about the distribution of actual states needs a stochastic transition law and evidence or proof about that law. A bell-shaped reference geometry cannot supply those missing ingredients.

36.4 An incompatible-reference extension, not the active model

To see why mass compatibility is load-bearing, temporarily allow $\sum_i x_i^* \neq M$ and ignore the nonnegative boundary while minimizing on the affine mass plane. A Lagrange multiplier λ gives

$$H(x - x^*) = \lambda \mathbf{1}, \quad x_i = x_i^* + \lambda \sigma_i^2, \quad \lambda = \frac{M - \sum_i x_i^*}{\sum_i \sigma_i^2}. \quad (36.4)$$

This candidate solves the nonnegative problem only if all its entries are nonnegative; otherwise active-boundary conditions are required. Even in the interior, the constrained optimum need not have $\mu_i = 0$: its marginals are equal, so conservative transfers have no first-order benefit.

The active examples do not use this extension. It is included to show why “zero gradient,” “constrained minimum” and “declared reference” are not universally interchangeable phrases.

Exercises and answers

At $(14, 6)$ with reference $(10, 10)$ and scales $(2, 2)$, the marginals are $(1, -1)$ and $V = 4$. At $(10, 10)$ both vanish. At $(2, 18)$ the potential is again sixteen, but the marginals reverse sign. Equal potential does not mean equal state or equal consequences for a directed action menu.

Explain why removing the old Allee term does not prove regeneration safe. The new closed lossless domain contains no such regenerative dynamics. The old theorem has not been refuted; its plant is simply not the active one.

Chapter recap

Gaussian geometry supplies a smooth, strictly convex quadratic and constant curvature. It makes several derivations exact and simple. It does not supply calibration, a probability law, a controller or a theorem of recovery.

Force, mobility, flux and finite action value

The potential describes a state. Its derivative describes how that potential changes in a direction. A constitutive response law describes how strongly a controller moves in response. A finite EBU quote describes the complete change caused by a specified action. These four statements fit together, but none should be substituted for another.

37.1 The force along a physical edge

For a lossless edge $i \rightarrow j$, the unit action column is $S_e = -e_i + e_j$, where e_i is the coordinate vector with one at position i . The directional derivative is

$$\mu^T S_e = -\mu_i + \mu_j. \quad (37.1)$$

Define the field force with the sign that makes a small improvement positive:

$$f_e = -S_e^T \mu = \mu_i - \mu_j. \quad (37.2)$$

For the Gaussian potential this is

$$f_e = \frac{x_i - x_i^*}{\sigma_i^2} - \frac{x_j - x_j^*}{\sigma_j^2}. \quad (37.3)$$

It compares standardized marginal deviations, not stocks alone. Two cells holding equal quantities can have different marginals if their references or scales differ. Conversely, unequal stocks can have equal marginals.

In the historical loss-aware edge model $S_e = -e_i + \eta_e e_j$, the same chain rule gives $f_e = \mu_i - \eta_e \mu_j$. The efficiency belongs inside the derivative because it changes the physical transformation. This general directional identity does not by itself adopt a loss-aware runtime or resolve its separate conservation and cost declarations.

37.2 Mobility and the Onsager-type response

The historical reference is Onsager's 1931 work on coupled irreversible processes [1, 2]. It connects linear force-rate relations and reciprocal coefficients

under microscopic-reversibility assumptions. EBU borrows a modelling structure, not an entitlement to those physical assumptions. A chosen deviation potential need not be free energy, and a declared mobility need not be an experimentally measured transport coefficient. This is why the qualifier “Onsager-type” matters.

The reason for introducing the structure is constructive: it separates the state’s driving slope from a mechanism’s responsiveness, and it permits an explicit conditional dissipation calculation. It does not follow from the finite accounting identity. We can change the response law while retaining that identity, but we cannot then retain the old response law’s dynamical proof.

The original EBU books studied the directed threshold-mobility law

$$J_e = M_e[f_e - \theta_e]_+, \quad M_e > 0, \quad \theta_e \geq 0, \quad (37.4)$$

where $[a]_+ = \max(a, 0)$. J_e is a rate. M_e is mobility: the strength of the declared response to force above threshold. In the present dimensionless potential normalization, M_e has units X^2/T and θ_e has units $1/X$. They are not the physical stock capacity or the persistent EBU capacity balance introduced later.

This is the project’s *Onsager-type* constitutive controller. The unthresholded signed relation $J = Mf$ is linear; the positive-part directed version is piecewise linear and blocks reverse flow on that edge. Neither form is derived merely by defining V . Choosing it is a separate modelling decision. The name does not establish universal thermodynamic validity for an economic system, nor does it imply a general coupled reciprocal-mobility matrix. The historical proofs concern their explicitly stated diagonal, directed controller.

If $M = 2$, $f = 1.4$ and $\theta = 0.5$, the requested rate is 1.8 in the corresponding units. Doubling mobility doubles this unconstrained request. It does not double the stock available at the source or authorize a physical cap to be exceeded. Permission may produce a different accepted rate, in which case a proof about the unconstrained law needs reinspection.

37.3 The exact finite quadratic

Let a declared action have finite increment δ . Expanding the square,

$$V(z + \delta) = \frac{1}{2}(z - x^* + \delta)^T H(z - x^* + \delta) \quad (37.5)$$

$$= V(z) + \mu(z)^T \delta + \frac{1}{2} \delta^T H \delta. \quad (37.6)$$

The two cross terms are equal because H is symmetric. Therefore

$$E(\delta) = V(z) - V(z + \delta) - C = -\mu(z)^T \delta - \frac{1}{2} \delta^T H \delta - C. \quad (37.7)$$

No small-action approximation was made. The first term is the tangent benefit; the second is the exact curvature correction. For $\delta = qS_e$ on a lossless edge,

$$E(q) = qf_e(z) - \frac{q^2}{2} (\sigma_i^{-2} + \sigma_j^{-2}) - C(q). \quad (37.8)$$

If $C = 0$ and $f_e > 0$, positive quantities give nonnegative value only up to $2f_e/(\sigma_i^{-2} + \sigma_j^{-2})$. Physical constraints can be tighter. The mathematical maximizer is half that upper zero, but the active random-affordable programme does not automatically choose the maximizer.

37.4 A force can be positive while a large action is harmful

At $z = (18, 2)$, $x^* = (10, 10)$ and $\sigma = (2, 2)$, the marginals are $(2, -2)$, so $f = 4$ and

$$E(q) = 4q - q^2/4. \quad (37.9)$$

At $q = 4$ the value is twelve. At $q = 8$ it is sixteen. At $q = 16$ it is zero, and at $q = 17$ it is $-17/4$. The initial force stays four in the frozen tangent calculation, but the field changes along the finite action. Quoting qf would ignore that change and overstate benefit.

The Onsager controller would supply a rate based on the force and its own mobility/threshold. A duration turns that rate into a quantity. Exact EBU then values the resulting accepted quantity. Keeping these steps explicit prevents the words force, flux and value from collapsing into one number.

Chapter recap

Force is a directional derivative; mobility is a constitutive response parameter; flux is the controller's rate; finite EBU is an endpoint difference. Chapter 44 derives the dissipation result for the Onsager-type law and explains exactly why it is not a recovery proof for a different actor.

Why local evaluation can be exact

The project asks for local valuation without a global observer deciding the worth of every action. The mathematical possibility comes from cancellation, not from approximating a large world by a small one. A local calculation is exact when it includes every potential term that can change.

38.1 The separable theorem

Let $V(x) = \sum_i v_i(x_i)$ and let W contain all coordinates changed by δ . For $i \notin W$, $z_i + \delta_i = z_i$. Split both sums:

$$V(z) - V(z + \delta) = \sum_{i \in W} [v_i(z_i) - v_i(z_i + \delta_i)] \quad (38.1)$$

$$+ \sum_{i \notin W} [v_i(z_i) - v_i(z_i)] \quad (38.2)$$

$$= V_W(z) - V_W(z + \delta). \quad (38.3)$$

No information about an unchanged term is needed to calculate its zero difference. This identity holds for nonlinear v_i and does not require Gaussian geometry. The Gaussian model is one especially transparent specialization of the historical locality result [4].

38.2 A complete small example

Use references (10, 10, 10) and scales (2, 2, 2) at state (18, 2, 12). The first two potential terms are eight each; the third is one-half. A four-unit transfer from cell one to cell two gives (14, 6, 12), with potential terms two, two and one-half. The global difference is $16.5 - 4.5 = 12$. The two-endpoint local calculation is $16 - 4 = 12$. The remote cell's value was not assumed zero. It cancelled because it was unchanged.

If an action also changes cell three, the proof demands its inclusion. Calling the third effect small or inconvenient does not permit its omission while retaining the adjective exact. One may define an approximation, but then its error and claim must be stated separately.

38.3 The factor version

A more general potential has factors

$$V(x) = \sum_{\alpha \in \mathcal{F}} \phi_{\alpha}(x_{A_{\alpha}}), \quad (38.4)$$

where A_{α} is the set of arguments used by factor α . Let $\mathcal{F}_W = \{\alpha : A_{\alpha} \cap W \neq \emptyset\}$. Every factor outside $\mathcal{F}_W$ cancels, so the exact difference is the sum over $\mathcal{F}_W$. A factor in that set may require reading a coordinate outside W . Write support and read support need not coincide.

For example a declared term $k(x_1 - x_3)^2$ changes when an action changes cell one. Cell three may remain physically untouched, yet its current value is required to evaluate the factor. A two-endpoint view that omits it cannot be called complete for that expanded potential. This is an illustrative extension, not an adopted nonseparable Gaussian model.

38.4 Three different locality claims

Valuation locality means the finite value needs only complete affected factors. *Information locality* means a particular actor or module may access only specified records. *Coordination locality* concerns how candidate groups are assembled and selected. A proof of the first does not establish the second or third.

A central program could enumerate every group in the world and evaluate each group locally. Its values would satisfy cancellation, but the selector would still be global. Similarly a class named `LocalView` is not a complete security proof if a helper can read unrestricted state elsewhere. Part III turns these distinctions into review questions without asserting that the future runtime already enforces them.

38.5 Frozen state and version completeness

Locality also requires agreement about time and parameters. Combining the source's present stock with the destination's stale stock does not evaluate one declared state. Changing a scale between the before and after terms does not compute one potential difference. Exact arithmetic cannot repair either mismatch.

Thus a reconstructible local quote binds the state version, affected factor identities, reference/scale version, action increment and process-burden declaration. Hashes can identify those inputs. They cannot certify that a sensor observed reality correctly; that is a measurement question.

Exercise and answer

Suppose a quote reports a difference of twelve while a global audit reports eleven. Name three possible causes before blaming numerical precision. An affected factor may be missing; the states may belong to different epochs; or one calculation may use a different parameter version. Each is a semantic discrepancy. The exact identity tells the reviewer where to look because, under its actual assumptions, such a difference is impossible.

Chapter recap

Local exactness is a cancellation theorem over complete affected factors. It is powerful precisely because its boundary is precise. It does not certify sensors, ownership, or decentralized selection.

From path integrals to committed quotes

A derivative tells us the effect of an infinitesimal displacement. A finite action travels through changing derivatives. Integrating them recovers the endpoint difference. This is the mathematical connection between force and exact EBU, and it remains valid whether or not a controller uses the force to choose the action.

39.1 One action, one chain rule

Let $\gamma(s) = z + s\delta$ for $0 \leq s \leq 1$, with V continuously differentiable along this path. The chain rule gives

$$\frac{d}{ds}V(\gamma(s)) = \nabla V(\gamma(s))^T \delta. \quad (39.1)$$

Integrate and change the sign:

$$V(z) - V(z + \delta) = - \int_0^1 \nabla V(z + s\delta)^T \delta \, ds. \quad (39.2)$$

For $\delta = qS_e$ with $q \geq 0$, substitution $r = sq$ yields $\int_0^q f_e(z + rS_e) \, dr$. The changing force, not just its initial value, is integrated. A declared process burden is then subtracted separately.

For the quadratic, $\mu(z + s\delta) = \mu(z) + sH\delta$. Integrating one and s over $[0, 1]$ immediately gives the finite expansion of Chapter 37. The average endpoint gradient is also exact:

$$V(z) - V(z + \delta) = -\frac{1}{2}[\mu(z) + \mu(z + \delta)]^T \delta. \quad (39.3)$$

This is not a general exact quadrature claim for arbitrary potentials. It works here because the gradient along the segment is affine.

39.2 Path arithmetic and path meaning

If the physical model declares constant-rate joint execution, s can parameterize its realized path after normalizing duration. If only endpoints are declared,

the same segment is an accounting interpolation. The endpoint identity is exact in both uses. Calling it an observed physical history requires the first additional declaration and, outside a simulation, appropriate measurement.

More generally, a smooth potential integrates to the same total along any admissible sufficiently regular path between the same endpoints. This does not imply equal physical cost for every route: process burdens and other state coordinates may differ. It also does not imply equal child allocations when a group path is decomposed in different ways.

39.3 A schedule is a function, not a promise about one quantity

A pre-action quote can commit $q \mapsto E(q)$ on a physically accepted domain. For the Ridge–Vale example the schedule is $4q - q^2/4$. If the accepted upper quantity is four and the measured quantity is three, the committed schedule gives $E(3) = 39/4$, not twelve. The measurement selects a point of an existing function; it does not authorize a new potential chosen after the outcome.

Quantity shortfall does not always mean a smaller signed value. On a decreasing part of a concave schedule, less movement can have greater value. The schedule, not a verbal monotonicity assumption, governs the account. If quantity exceeds the committed domain, the old positive quote cannot simply be extended. The historical foundation retains an explicit overexecution boundary rather than inventing punitive physics.

39.4 Convexity and activation burden

For a convex differentiable V , $V(z + \delta) \geq V(z) + \nabla V(z)^T \delta$. Hence exact value is no greater than its tangent value when both subtract the same C . For the positive definite quadratic the inequality is strict for nonzero δ .

Along a fixed edge, the field-benefit schedule is concave. The complete schedule remains concave when the declared $C(q)$ is convex. Nonnegative cost alone is not sufficient. A fixed activation cost with $C(0) = 0$ and $C(q) = c_0 > 0$ for $q > 0$ creates a downward jump at zero; interpolation across it would invent an unregistered discount. The first prospective lossless domain avoids this complication by declaring $C = 0$, not by claiming real operations are universally costless.

39.5 What a quote does not guarantee

A valid schedule does not establish permission, ownership, sensor integrity, or execution. A positive quote does not prove the action will occur. A negative

quote does not prove that essential service should be denied. The mathematics describes represented field effect. Institutions and the experimental capacity policy are additional objects with additional claims.

Chapter recap

The exact finite difference and the integrated force are two expressions for one field change. A committed schedule binds that expression to a baseline, rule and domain. Physical-path meaning and institutional consequences do not follow from integration alone.

Sequences, cycles and external change

The simplest accounting theorem in EBU is also one of the most important. If every action begins where the previous one ended, the intermediate potentials cancel. A surprisingly large number of incorrect issuance rules fail because they stop using those live intermediate states.

40.1 Undriven telescoping

Let $x^0, x^1, \dots, x^N$ be a declared sequence under one fixed potential, with action values $E_k = V(x^k) - V(x^{k+1}) - C_k$. Then

$$\sum_{k=0}^{N-1} E_k = V(x^0) - V(x^N) - \sum_{k=0}^{N-1} C_k. \quad (40.1)$$

To prove it, write the first two differences: $V(x^0) - V(x^1)$ and $V(x^1) - V(x^2)$. Their shared term cancels. Repeating that cancellation leaves only the first and last endpoints. Neither convexity nor Gaussian geometry is required for this algebraic identity.

For a closed physical cycle $x^N = x^0$, the total is $-\sum C_k$. If all burdens are nonnegative, it cannot be positive. With $C = 0$ it is zero, not necessarily negative. A reversible movement can return every represented coordinate to its start without making the account grow.

40.2 A live-state calculation

Use (18, 2) with reference (10, 10) and scales (2, 2). Transfer two units, then three, then one. The successive potentials are

$$16, \quad 9, \quad 9/4, \quad 1. \quad (40.2)$$

The action values are 7, $27/4$ and $5/4$, whose sum is fifteen. That equals $16 - 1$. Quoting all three against the first state instead would replace the last two live values with unrelated counterfactuals.

Splitting a transfer preserves the field total when the resulting sequence has the same endpoint. It need not preserve separate activation costs, physical delays or intervening disturbances. A theorem about endpoint differences is not a theorem that every way of organizing work costs the same.

40.3 Forcing before the action epoch

The new programme proposes an external change from x_t to z_t before actors evaluate their candidates. Define

$$D_{\text{ext},t} = V(z_t) - V(x_t), \quad E_t = V(z_t) - V(x_{t+1}) - C_t. \quad (40.3)$$

Add and subtract $V(x_t)$ in the second expression and sum:

$$\sum_{t=0}^{N-1} E_t = V(x_0) - V(x_N) + \sum_{t=0}^{N-1} D_{\text{ext},t} - \sum_{t=0}^{N-1} C_t. \quad (40.4)$$

An actor who does nothing has zero field value at the shared post-forcing baseline. Nature's own change appears in D_{ext} , not in an idle actor's income. The external term can be negative: forcing may move the system closer to the reference.

The old D0 foundation uses a frozen-drive no-action successor. Its immediate cancellation reasoning is related, but its event order is not identical to this proposed forcing-first profile. An implementation needs an explicit profile contract rather than renaming one variable and claiming equivalence.

40.4 A forcing example with exact signs

Begin at reference (10, 10), where $V = 0$. A conservative external shift produces (14, 6), where $V = 4$. An actor then transfers two units back, ending at (12, 8) with $V = 1$. The external audit is +4 and the actor's field value is +3. The total physical potential change is only +1; confusing it with actor value would get both the baseline and the sign wrong.

If a later external shift takes (12, 8) to (10, 10), its audit is -1 . No actor performed that improvement. Crediting it to the next idle actor would violate the declared separation even if a global sum happened to close.

40.5 Changing the potential is another accounted event

If parameters vary, simple telescoping under one V no longer applies. Writing V_t for the version at event t , parameter changes create terms such as $V_{t+1}(x) - V_t(x)$ even at unchanged physical state. A recalibration can therefore create apparent value unless its effect is separated from physical actions. The first model freezes references and scales precisely to avoid silently mixing those effects.

The historical no-overissue theorem retains its original one-action and audit assumptions. This chapter derives a general endpoint identity; it does not

retroactively broaden that source theorem's settlement scope. For simultaneous groups, the joint transition must be valued once and any division needs the distinct treatment in the next chapters.

Chapter recap

Live endpoints, fixed parameters and complete external accounting make the sum reconstructible. Closed-cycle closure rules out one accounting exploit. It does not prove that actors prefer recovery over cycling or refusal.

Groups, cross terms and honest comparisons

Two actors can share a source without either actor's request being wrong on its own. The problem is that both requests refer to one source, not two copies of it. Nonlinear valuation must therefore evaluate their joint change before asking how any resulting value might be divided.

41.1 One group, one endpoint

For fixed accepted increments δ_a , define $\delta_G = \sum_{a \in G} \delta_a$. In the cost-free Gaussian domain,

$$E_G = -\mu(z)^T \delta_G - \frac{1}{2} \delta_G^T H \delta_G. \quad (41.1)$$

Expand the quadratic sum. Symmetry of H gives

$$E_G = \sum_{a \in G} E_a^{(0)} - \sum_{a < b} \delta_a^T H \delta_b, \quad (41.2)$$

where $E_a^{(0)} = V(z) - V(z + \delta_a)$ is the hypothetical value of child a alone at the same baseline. These singleton values are not sequential receipts. The cross terms are exactly what their naive sum leaves out.

41.2 A shared-source calculation

At $(18, 2)$ with the usual reference and scales, let two children each send two units along the same edge. Each alone has value seven. Together they send four and have value twelve, not fourteen. Their increments are both $(-2, 2)$, so $\delta_A^T H \delta_B = 2$ and the correction is -2 .

The same issue occurs when the destinations differ. If two children share only a source with quantities $a, b > 0$, their cross term contains $ab/\sigma_i^2 > 0$. The joint field value is smaller than the sum of their separate source-relief claims by that amount. Disjoint changed coordinates have zero cross term for the diagonal model.

Opposing increments can instead produce a positive correction. Take $\delta_A = (-2, 2)$ and $\delta_B = (2, -2)$ at the same baseline. Their standalone values are seven and minus nine. The net increment is zero, so the group value is zero

and the correction is $+2$. Positive nonadditivity here means cancellation of represented displacement. It is not proof that real physical circulation is free or socially beneficial.

41.3 Same-baseline nonadditivity is not a parallel advantage

Define $N_G = E_G - \sum_a E_a^{(0)}$. Now choose an admissible sequential schedule π , with complete endpoint x_π . Its field total is $V(z) - V(x_\pi)$. The comparator-relative difference is

$$I_{G|\pi} = E_G - E_\pi = V(x_\pi) - V(z + \delta_G). \quad (41.3)$$

When fixed additive increments can be serialized feasibly, the final endpoint is the same and $I_{G|\pi} = 0$. Yet N_G may be nonzero. The shared two-unit example has $N_G = -2$ and $I = 0$: it contains nonlinear same-baseline double counting but no endpoint advantage of simultaneity.

A resolver replay can change this comparison. If it recomputes accepted quantities after the first action, it may produce another endpoint. That is a different, explicitly named comparator, not evidence that the same accepted vector has two contradictory values. The bridge source [5] distinguishes quantity-fixed from rule-replayed comparisons.

41.4 The conditional Boolean-lattice calculation

For a set A of fixed children, let $F(A) = V(z) - V(z + \sum_{a \in A} \delta_a)$ and $F(\emptyset) = 0$. The pair inclusion–exclusion coefficient is

$$m_{ab} = F(\{a, b\}) - F(\{a\}) - F(\{b\}) = -\delta_a^T H \delta_b. \quad (41.4)$$

Because the indicator expansion of F has degree at most two, all coefficients of order three and higher vanish. This statement needs the same baseline, potential and fixed increments in every subset.

If a subset reruns a resolver, its increments can differ. The resulting set function need not be quadratic in membership indicators. Furthermore, physically inadmissible subsets cannot be turned into valid interventions by algebraic convenience. Every relevant subset must be admitted under the declared protocol before a physical Boolean-lattice interpretation is made. Möbius analysis remains an optional diagnostic, not extra issuance.

Questions to ask of a group claim

Were the children valued at one baseline or consecutive states? Were quantities held fixed or re-resolved? Is the comparator physically admissible? Does

it include the same costs and complete state? Is a correction being mistaken for a causal contribution? These questions determine the meaning of an interaction number before its sign is interpreted.

Chapter recap

The quadratic cross term explains why independent same-baseline values fail to close on a shared group. A named sequential comparator answers another question. Neither calculation supplies an ownership rule.

A path attribution is not an entitlement

Once the group value is known, one may want a number beside each child action. A decomposition can be mathematically exact without being a uniquely identified causal contribution. The difference must be kept visible before any such number is connected to a persistent account.

42.1 The declared common-path decomposition

Choose the straight group path $\gamma(s) = z + s\delta_G$. Define

$$R_a = - \int_0^1 \mu(z + s\delta_G)^T \delta_a ds = -\mu(z)^T \delta_a - \frac{1}{2} \delta_a^T H \delta_G. \quad (42.1)$$

The last equality uses the affine Gaussian gradient. Sum over children:

$$\sum_a R_a = - \int_0^1 \mu(z + s\delta_G)^T \sum_a \delta_a ds \quad (42.2)$$

$$= - \int_0^1 \mu(z + s\delta_G)^T \delta_G ds = E_G. \quad (42.3)$$

This proves closure for the cost-free domain. If $C_G \neq 0$, the path terms still sum to the field component; allocating or accounting for cost requires its own declaration. It must not disappear in the notation.

42.2 What the formula assigns in a small group

Let the children send one and three units from Ridge to Vale at $(18, 2)$. The group increment is $(-4, 4)$ and its value is twelve. Initial tangent benefits are four and twelve. Curvature corrections are one and three, so $R_A = 3$, $R_B = 9$. The total closes exactly.

Their standalone values are $15/4$ and $39/4$, summing to $27/2$. Their sequential live values depend on order. Sending one first gives $15/4$ followed by $33/4$; sending three first gives $39/4$ followed by $9/4$. Both sequential sums equal twelve. Neither ordered split is the same as $(3, 9)$, because it describes another allocation convention.

The sum identity cannot tell us which of these child descriptions is fair, deserved or causally identified. It tells us which declared arithmetic is being used and whether that arithmetic closes on the group.

42.3 Physical path versus interpolation

Under a declared constant-rate physical model, each child's normalized rate contributes δ_a along the common path. The integral then decomposes that model path. Under endpoint interpolation it is an accounting construction. Identical arithmetic does not erase the distinction.

Historical sources use the name radial Aumann–Shapley for related path allocation [4]. Using or avoiding that name does not resolve the interpretation. The scientifically useful record specifies the actual path and assumptions, labels the result as a mathematical attribution, and states what it does not establish: ownership, fairness, causal entitlement, or permission to spend.

42.4 Splitting and cancellation

Suppose one child is replaced by two proportional subchildren with increments $\lambda\delta_a$ and $(1 - \lambda)\delta_a$, while the group path is unchanged. The formula is linear in the child increment, so their attributed values add to the original R_a . This is a limited mathematical splitting property. A split that changes grouping, fees, ownership or feasibility is not covered by it.

If $\delta_G = 0$, the formula becomes $R_a = -\mu(z)^T \delta_a$. These child values can be nonzero and cancel. In the opposing two-unit example they are +8 and −8, while the group has zero value. This illustration makes the interpretive risk vivid: a cancelling endpoint does not establish a gift of eight units to one participant and a legitimate charge of eight to another.

42.5 From children to owners is another map

If a prospective policy assigns every child a unique owner, then finite-sum rearrangement gives

$$\Delta B_i = \sum_{a: \text{owner}(a)=i} R_a, \quad \sum_i \Delta B_i = \sum_a R_a. \quad (42.4)$$

The second equality is exact once the map exists. It does not select the map. A requester, provider, beneficiary and physical source can be different entities. Substituting one for another because a software field is convenient would change the policy while leaving the arithmetic looking correct.

Chapter recap

Common-path attribution provides an explicit, auditable decomposition. Its proof closes a sum. Ownership and spendability require separate prospective authority even for a group with only one child.

Capacity accounts and random affordable action

The new research question is not whether a controller can be written to move downhill. The Onsager-type example already shows how a particular response law can have a descent property. The proposed question is what happens when actors choose without ranking EBU, but their choices are restricted by a separately declared capacity account.

43.1 The policy, stated conditionally

Let $B_i \geq 0$ be persistent experimental capacity for owner i , initially zero. It is not physical stock, installed pumping capacity or a real-world monetary entitlement. Suppose an explicitly adopted owner/path policy would assign group changes ΔB_i satisfying

$$\sum_i \Delta B_i = E_G. \quad (43.1)$$

The proposed affordability condition is

$$B_i + \Delta B_i \geq 0 \quad \text{for every affected owner } i. \quad (43.2)$$

The intended policy permits same-event signed netting within one owner, not across all owners. It introduces no loan, refill or global pool. Its adoption is a modelling choice still requiring the programme's declared authority, not a consequence of Chapter 42.

43.2 Why positive group value is insufficient

Consider a candidate whose owner changes are $(3, -1)$ and whose aggregate value is two. With zero balances, it is unaffordable for the second owner. Positive total value does not cover that owner's debit unless a different policy explicitly permits pooling. Conversely, if one owner holds both child actions, same-owner netting would produce $+2$ and pass. Ownership can therefore change the feasible stochastic process without changing the physical endpoint or group field value.

With positive existing capacity, a negative-valued action may be affordable. That is not an accounting error. It is one intended reason this policy is not a monotone potential-descent controller. Whether spending later enables useful action or merely produces cycles is a research question.

43.3 The external-audit invariant

Let J accumulate $D_{\text{ext},t} = V(z_t) - V(x_t)$ and suppose the adopted cost-free updates are exact. Then

$$J_{t+1} = J_t + D_{\text{ext},t}, \quad (43.3)$$

$$\sum_i B_{i,t+1} = \sum_i B_{i,t} + V(z_t) - V(x_{t+1}). \quad (43.4)$$

Adding terms yields

$$V(x_{t+1}) + \sum_i B_{i,t+1} - J_{t+1} = V(x_t) + \sum_i B_{i,t} - J_t = K. \quad (43.5)$$

At reference with zero genesis and zero audit, $K = 0$. Forcing does not directly fund an actor. Its signed potential change enters J so that the full account can be reconstructed.

After a single shock, suppose $J = 4$ and $K = 0$. If an actor reduces potential to one, total capacity becomes three. Later an affordable negative action could raise potential to two while lowering total capacity to two. The identity permits this. It gives a constraint, not a ranking of choices.

43.4 What can be proved about the reference

At $x = x^*$, any nonzero net increment has $E_G = -\frac{1}{2}\delta_G^T H \delta_G < 0$. At zero balances, owner-wise affordability would require every $\Delta B_i \geq 0$, making their sum nonnegative. Thus no nonzero-net action can be affordable there under the stated cost-free policy. A cancelling group may have zero aggregate change; its owner-level feasibility still depends on the declared attribution/map.

This is a conditional statement about zero genesis at the reference. After a shock and earned capacity, the account can permit uphill movement. Therefore the zero-genesis observation does not establish attraction from disturbed states or long-run recovery.

43.5 Random choice is not mathematical neutrality

“Choose uniformly” is incomplete until the sample space is fixed. Uniform choice among edges differs from uniform choice among quantities, groups or

owners. If one physical option appears twice under different identifiers, sampling entries gives it twice the probability. A finite menu is also not the same as the continuum of all physically possible quantities.

The programme intends EBU-blind candidate construction and unweighted choice among affordable physically feasible options, with a physical-random control that lacks the affordability filter. Exact menu, duplicate, no-action and group-assembly semantics remain load-bearing decisions. They must not be chosen after inspecting which version makes EBU look better.

Chapter recap

Capacity and external audit can satisfy an exact conditional invariant. That invariant does not make the policy fair, decentralized, monotone or successful. Those are separate statements about the declared process.

Onsager-type dissipation and finite-step limits

The force–mobility–flux structure deserves its own proof, both because it motivated earlier EBU models and because it shows how much a dynamical conclusion depends on the chosen response law. We derive the result without calling every Gaussian actor an Onsager controller.

44.1 The continuous-time calculation

Suppose, on a time interval where the declared physical model is valid,

$$\dot{x} = u(x) + SJ(x), \quad f = -S^T \mu, \quad J_e = M_e[f_e - \theta_e]_+. \quad (44.1)$$

The chain rule gives

$$\dot{V} = \mu^T u + \mu^T SJ = \mu^T u - \sum_e f_e J_e. \quad (44.2)$$

For an active edge, $J_e = M_e(f_e - \theta_e) > 0$, so $f_e = J_e/M_e + \theta_e$. For an inactive edge both J_e and its product with f_e vanish. Consequently

$$\dot{V} = \mu^T u - \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right). \quad (44.3)$$

With $u = 0$, $M_e > 0$ and $\theta_e \geq 0$, every subtracted term is nonnegative and $\dot{V} \leq 0$. This is a theorem about this declared continuous-time law. It does not require the historical hinge potential; the chain rule works for the smooth Gaussian potential too.

44.2 What the proof did not check

Nothing in the sign calculation alone ensures nonnegative stock, physical capacity, or existence of a trajectory for all time. A source could be asked to send beyond its available stock. A later cap or projection may repair feasibility but changes the vector field. One must then verify the actual accepted flux's sign/constraint properties, not cite the unmodified equality.

External drive contributes $\mu^T u$ and can raise the potential. Vanishing flux also need not mean $x = x^*$. Directed edges, positive activation thresholds, disconnected components and physical constraints can create other stationary configurations. A nonincreasing scalar is not by itself a proof of convergence to its unique unconstrained minimizer.

44.3 The finite Euler update

Freeze $w = u + SJ$ and take $x^+ = x + \Delta t w$. For the Gaussian model,

$$V(x^+) - V(x) = \Delta t \mu^T w + \frac{1}{2} \Delta t^2 w^T H w. \quad (44.4)$$

The second term is nonnegative and exact. Continuous-time dissipation can therefore coexist with an overshooting finite step. Put $D = -\mu^T w$. If $D > 0$ and $w \neq 0$, a sufficient and in this quadratic case exact condition for nonincrease along that frozen direction is

$$0 < \Delta t \leq \frac{2D}{w^T H w}. \quad (44.5)$$

This does not certify physical feasibility. If $D = 0$ and $w \neq 0$, every positive step raises V ; if $w = 0$, the state does not change.

44.4 A complete controller example

At $(18, 2)$, use one edge, $M_e = 1/2$, $\theta_e = 0$ and $u = 0$. Then $f = 4$, $J = 2$ and $w = (-2, 2)$. We have $\mu^T w = -8$ and $w^T H w = 2$, giving

$$V(x^+) - V(x) = -8\Delta t + \Delta t^2. \quad (44.6)$$

The nonincrease interval is $0 \leq \Delta t \leq 8$. At duration four, the quantity is eight and the reference is reached in this particular two-cell frozen step. At duration eight, the endpoint is $(2, 18)$ and potential returns to sixteen. At duration nine, the physical endpoint $(0, 20)$ is still nonnegative but has potential twenty-five. Continuous descent did not license an arbitrarily large Euler step.

These are analytic substitutions into one frozen update, not a simulated time series or parameter search. Their role is to expose the curvature term.

44.5 Network geometry and a general bound

For a full vector of edge rates, the relevant quadratic is $J^T S^T H S J$. Shared nodes produce off-diagonal terms in $S^T H S$ even when H is diagonal. Thus a certificate derived by treating every edge in isolation can miss joint curvature.

The Gaussian gradient is globally Lipschitz with constant $L = h_{\max}$. Indeed $\mu(x) - \mu(y) = H(x - y)$ and the diagonal operator norm of H is $h_{\max}$. Therefore $w^T H w \leq L \|w\|_2^2$, yielding the familiar sufficient descent-lemma bound. The original network and timestep proofs are preserved in Appendix H-II, but their old numerical constants must not be copied over the new Hessian or a different actor law.

44.6 Why keep Onsager in the books?

It supplies a transparent benchmark of reasoning: declare a response, derive its sign structure, check feasibility, then inspect discretization. It also explains the origin of mobility and why force is not itself a flux. What it does not supply is a hidden restoring plant for the new programme. Random affordable actors can spend capacity on negative-valued actions; their increments are not generally $\Delta t S J(f)$. Applying the theorem to them would change the subject of the experiment.

Chapter recap

Onsager-type dissipation is retained as conditional mathematics and an explicit controller alternative. Exact finite EBU survives controller changes; the old controller's dynamical guarantees do not automatically do so.

Feasibility, routes and the long-run boundary

An action account can be exact while a world is unable to meet its needs. A potential can have a unique minimum while a process never reaches it. These are not paradoxes. They follow from the distinction between a landscape, a physical domain and a transition rule.

45.1 The historical preservation certificate

P1C addresses a specific regenerative-source export question. In its historical applicable state and typing, a robust export-rate budget has the form

$$Q_{\max} = \frac{[(x - \varepsilon_x) + \Delta t(u - \varepsilon_u) - R_{\text{eff}}]_+}{\Delta t}. \quad (45.1)$$

R_{eff} is a stock quantity. It is not the old homeostatic reserve R , the Gaussian scale, or the new capacity balance B . Under the certificate's feasibility and uncertainty assumptions, capping aggregate export protects its stated successor boundary. It does not move material toward a destination that lacks it.

The source classifier matters as much as this arithmetic. The preserved P1C design does not grant preservation-safe export from every finite stock merely because the expression is positive. Its regenerative-source premise cannot be removed to create a permission law for the closed finite Gaussian world. Appendix H-II retains the original proof at its own scope.

45.2 A route is a sequence of physical transformations

Take a three-cell teaching state (18, 10, 2), reference (10, 10, 10), and equal scales two. Its potential is sixteen. Move four units from Ridge to the hub: the endpoint is (14, 14, 2) and the potential is twelve. Then move four from the hub to Vale: (14, 10, 6) has potential four. The first edge contributes four and the second eight, totaling twelve.

The intermediate hub matters even though its final stock is unchanged. A direct transfer, if it existed, would have the same field endpoint value in this lossless example, but need not have the same physical costs, timing or permissions in a fuller world. One cannot erase an intermediate fuel, electricity or storage transformation simply because endpoint algebra is convenient.

45.3 Planning is not local valuation

A planner might ask which sequence of actions reaches a target at minimum cost. Bellman reasoning compares present choices and future continuations. That can be a legitimate separately declared problem, but it introduces state, horizon, objective and information requirements beyond a present local quote. A random-affordable actor must not secretly call that planner and still be described as choosing without target-directed optimization.

There is also a difference between a route's total favorable field value and its affordability at every intermediate point. A useful route can begin with a negative-valued action. End-to-end positivity does not fund that first step under zero balances without an explicitly different policy. This is a possible limitation to report, not a reason to add silent borrowing.

45.4 Boundedness is built into the closed domain

Since $0 \leq x_i \leq M$,

$$V(x) \leq \frac{1}{2} \sum_i \frac{\max\{(x_i^*)^2, (M - x_i^*)^2\}}{\sigma_i^2} < \infty. \quad (45.2)$$

Both an EBU-filtered arm and a physical-random control inherit this bound if they stay in the same simplex. Showing that one arm's potential remains finite cannot discriminate their usefulness. A long-run study must ask about quantities that can genuinely differ: distance occupation, recovery time, cycling, refusal, distribution or declared service.

45.5 Invariance, stability and attraction

An invariant domain contains every allowed successor. Stability of a reference means sufficiently small departures remain small under the specified dynamics. Attraction means departures approach the reference. Stationarity is a property of a probability law under a specified transition kernel. These statements answer different questions.

Strict convexity alone proves none of them for an arbitrary actor process. A finite menu can trap the system away from reference; a capacity policy can permit cycling; a directed graph can block useful routes. Even if the physical state has a stationary distribution under continuing forcing, the audit J and balances can drift. A stationary claim about the augmented process (x, B, J) needs its own argument.

An adverse case worth keeping

Consider two cells with odd integer stocks and only even transfer quantities. If the reference requires even stocks, every reachable stock remains odd: adding or subtracting an even number preserves parity. The reference can be mass-compatible but unreachable under that menu. This exact obstruction is not a Gaussian calibration result and should not be removed after a study reveals it. It is a reason to state the menu before testing recovery.

Chapter recap

Permission, route reachability and long-run dynamics are separate from exact valuation. Keeping them separate makes negative results intelligible rather than embarrassing exceptions to be hidden.

A complete Gaussian field laboratory

This laboratory is a sequence of exact worked calculations. It is not a trajectory produced by the research engine and not a proposed configuration for a scientific campaign. Its purpose is to let the reader reconstruct every important distinction with pencil-and-paper arithmetic.

46.1 Write the declaration before calculating

For the first cases, use two cells with

$$z = (18, 2), \quad x^* = (10, 10), \quad \sigma = (2, 2), \quad H = \text{diag}(1/4, 1/4). \quad (46.1)$$

Both state and reference sum to twenty. The action is a lossless transfer from the first cell to the second, with finite quantity q . The increment is $(-q, q)$. The illustrative physical domain permits $0 \leq q \leq 18$; there is no separate destination cap in this example. Process burden is zero.

The initial potential is

$$V(z) = \frac{1}{2}[(8/2)^2 + (-8/2)^2] = 16. \quad (46.2)$$

The marginals are $(2, -2)$, so the field force is four. Substitution into the exact finite expression gives $E(q) = 4q - q^2/4$. Every later number should be checked both from that polynomial and from the endpoint potential.

46.2 A menu with favorable and adverse quantities

q	Endpoint	V after	$E(q)$
0	(18, 2)	16	0
1	(17, 3)	49/4	15/4
2	(16, 4)	9	7
4	(14, 6)	4	12
8	(10, 10)	0	16
12	(6, 14)	4	12
16	(2, 18)	16	0
17	(1, 19)	81/4	-17/4
18	(0, 20)	25	-9

Each row conserves twenty and has nonnegative stocks. Therefore physical feasibility alone does not choose the helpful rows. Conversely, the two zero-valued quantities mean very different things: $q = 0$ does nothing, while $q = 16$ exchanges the deviations and returns to the same potential. Equal potential is not equal state.

The table is not a controller instruction. Selecting eight because it has the largest value would define an optimizer. Choosing uniformly from a declared affordable subset would define another process. Listing analytical options does not authorize either one as the scientific study.

46.3 Check the integral independently

Along the transfer, $f(r) = 4 - r/2$. For $q = 4$,

$$\int_0^4 (4 - r/2) dr = [4r - r^2/4]_0^4 = 12. \quad (46.3)$$

The initial-tangent estimate is sixteen, overvaluing the action by four. The average force along the segment is $(4 + 2)/2 = 3$, and multiplying by quantity four again gives twelve. The endpoint-force averaging is exact because this particular gradient is affine, not because trapezoidal quadrature is exact for all field models.

At $q = 12$, the final force is minus two. The average is one and the integral is twelve. Some portions of the path improve the potential and later portions worsen it, yet the total remains positive. A positive integral is not a statement that every infinitesimal part was beneficial.

46.4 Separate force from an Onsager response

Declare, only for this illustration, $M_e = 1/2$ and $\theta_e = 0$. At the original state the requested rate is two. Duration $1/2$ gives quantity one and exact field value $15/4$. Duration two gives quantity four and value twelve. Duration four gives quantity eight and value sixteen. These are three frozen-step calculations, not repeated steps in one run.

If instead the flux were recalculated continuously as the state moved, the force would fall and so would the flux. It would be a mistake to interpret the frozen duration-four jump as the solution of that differential equation. “Same starting force” does not mean “same time evolution.”

46.5 Local evaluation with a nonzero remote term

Append a third cell at stock twelve, reference ten and scale two, but do not change it. The initial global potential becomes $33/2$. After the four-unit

transfer it is $9/2$. The remote half-unit contribution is present in both and cancels. Local valuation still gives twelve.

Now imagine the transfer also consumes stock from that third cell. The former two-cell formula is no longer a complete local account. The example does not need a numerical error to demonstrate the problem: its support assumption has changed. A correct implementation should reject the incomplete declaration rather than silently use the old result.

46.6 Sequential receipts and reversed order

Divide a four-unit transfer into quantities one and three. If one occurs first, its value is $15/4$ and the subsequent three-unit value is $49/4 - 4 = 33/4$. If three occurs first, its value is $39/4$ and the last one-unit value is $25/4 - 4 = 9/4$. Both sums are twelve.

The intermediate state explains the different child values. There is no contradiction to reconcile by averaging the orders unless a separate allocation rule asks for such an average. The physical sequence already defines which live state each action used.

If an extra nonnegative activation burden is incurred for each separate action, splitting can change the net total. The field sum still telescopes; the process-cost sum changes. An implementation must not quote the one-action cost and then execute two independently activated operations under it.

46.7 A simultaneous group on the same source

Let both two-unit children be accepted on the same baseline. The net increment is $(-4, 4)$ and the group field value is twelve. Same-baseline singletons each show seven, yielding a misleading fourteen if added as though independent. The missing cross correction is minus two.

Under the declared straight common path, the attribution to each child is six. This is easy to check from the average group gradient, or by giving each child half of the identical net direction. Closure is $6 + 6 = 12$. It does not prove that each actor owns six units of spendable capacity.

For unequal children of one and three units, common-path values become three and nine. These differ from both ordered sequential splits above. The group field result is the same because the endpoint is the same; the child interpretations refer to different declared constructions.

46.8 Cancellation and a signed child account

Use the same baseline but take opposing children $(-2, 2)$ and $(2, -2)$. The net endpoint is unchanged. Standalone values are 7 and -9 , while common-path values are 8 and -8 . Both the group value and the sum of path values are zero. The singleton sum is minus two.

If each child belongs to a different zero-balance owner, the proposed owner-wise affordability rule refuses the owner charged eight. If both belong to one owner, same-event netting gives zero. This is an explicit demonstration that ownership policy can matter even when the physical net change is identical. It is not a recommendation to choose the ownership map that produces a desired outcome.

46.9 A different scale is a different model

Keep the original state and reference but change scales to $(2, 4)$. The initial potential is ten, marginals are $(2, -1/2)$, and field is $5/2$. For quantity four the curvature coefficient is $\frac{1}{2}(1/4 + 1/16) = 5/32$, so

$$E(4) = 4(5/2) - (5/32)16 = 15/2. \quad (46.4)$$

The endpoint potential is $5/2$, agreeing with $10 - 5/2 = 15/2$. Nothing physical changed between this declaration and the earlier example. The change in value comes from the declared scale at the second cell. Any empirical use must therefore justify and version that declaration.

Changing to smaller physical units would instead rescale stock, reference, quantity and scale together and leave EBU unchanged. A field account that changes when only the unit label changes has a dimensional bug.

46.10 A route whose first step is unfavorable

Declare a new three-cell example with state $(10, 10, 2)$, reference $(6, 6, 10)$ and equal scales two. Both totals are twenty-two. The initial potential is twelve. A four-unit transfer from the first cell to the second gives $(6, 14, 2)$ with potential sixteen, so its value is minus four. A subsequent eight-unit transfer from the second cell to the third reaches the reference $(6, 6, 10)$ and has value sixteen. The route total is twelve.

This is a physically feasible quantity-fixed sequence under simple stock non-negativity. It illustrates the difference between favorable endpoint value and intermediate affordability. A zero-balance owner cannot pay the first negative action under the proposed no-borrowing rule. The later positive action does not retroactively authorize it. A different grouping, timing or ownership policy would be a different declared experiment.

46.11 Forcing and the complete account

Begin instead at $(10, 10)$ with $V = 0$, zero balances and zero external audit. A conservative shock gives $(14, 6)$, with $D_{\text{ext}} = 4$. One two-unit restoring transfer ends at $(12, 8)$, with potential one and field value three. If the candidate account policy credits that action to its declared owner, the invariant reads

$$1 + 3 - 4 = 0. \quad (46.5)$$

Now consider a possible reverse two-unit action from $(12, 8)$ back to $(14, 6)$. Its value is minus three. An owner with balance three can afford it, returning to potential four and balance zero, still with audit four. Both accounts close. Recovery has been undone. This is a counterexample to the claim that the accounting invariant alone forces monotone recovery.

No random process has been run to obtain this pair of possibilities. It is an algebraic construction showing what the proposed rules allow. A later study must determine how often such events arise under its actual menus and sampling, without changing those rules after seeing results.

46.12 A reproducible hand calculation

For every candidate, a reader should be able to list the baseline, references, scales, child increments, complete support, physical constraints, before/after potential, group value and optional path values. If capacity is discussed, the owner map and prior balances must also be listed. If forcing occurs, the before-forcing state and signed audit must be listed separately.

An unexplained number in any one of those positions should interrupt the claim. The appropriate response is not to fill it with a plausible value. It is to identify which declaration, measurement or authority is missing. This is how a careful laboratory connects to the fail-closed software architecture described in Part III.

Final synthesis

The Gaussian model gives exact, local finite accounting and a transparent curvature correction. It supports conditional path and telescoping proofs, and it makes group interaction calculable. The Onsager-type controller gives an additional conditional dissipation law when it is explicitly chosen. Random-affordable action is a different prospective process. Its useful, adverse and inconclusive outcomes remain to be investigated, not written into the definition of success.

Sources and authority map

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How to verify a source claim

The companion source manifest records full file hashes, the checkout identity, the original PDF identities and the exact original pages retained in each appendix. The outline disposition records subjects rewritten rather than claiming every original sentence survives. All new derivations state their assumptions in the chapter itself. A citation identifies a source; it does not transform a prospective design into an accepted theorem or experiment.

Historical Appendix H-II

The following thirty pages preserve original Part II Chapters 38–41, original PDF pages 65–94. They derive continuous-time dissipation, numerical step restrictions, network bounds and the P1C one-step preservation result. Original body text, equations and printed folios are retained; an added margin banner identifies every page as historical material. Chapter titles are displayed smaller to match the author’s typography correction.

These are useful conditional arguments, not the active Gaussian actor definition. In particular, the automatic threshold-mobility law is not random-affordable selection; the old hinge curvature constants are not the Gaussian Hessian; and a P1C regenerative-source export certificate is not a permission law for closed internal redistribution. The active Chapters 44–45 explain these distinctions and rederive the applicable Gaussian quantities.

Original cross-references on these pages refer to the original edition. Consult the preserved original PDF when following one outside this extract. The old end-to-end laboratory remains in that original, while Chapter 46 of this edition supplies a new Gaussian laboratory rather than altering old calculated outcomes. No result in this appendix is new empirical evidence.

Continuous-time dissipation, derived line by line

So far we have valued finite actor actions. We now study a related physical controller: a continuous edge flow generated by the local force. The purpose of this chapter is narrow and important. It proves that, in the absence of external drive, this particular flow law cannot increase the declared potential.

The result is sometimes described in one line as “the field dissipates”. We will not leave it at that. We will derive every term, check its unit, and show why the theorem does not apply to arbitrary circulation or arbitrary actor behaviour.

38.1 Continuous time and the dot notation

Let $x(t)$ denote the state at continuous time t . A dot above a variable means derivative with respect to time:

$$\dot{x}(t) = \frac{dx}{dt}(t). \quad (38.1)$$

If x has unit X , then $\dot{x}$ has unit X/T . The continuous network law is

$$\dot{x} = u(x) + SJ(x). \quad (38.2)$$

The vector $u(x)$ contains natural or imposed rates at nodes. The vector $J(x)$ contains edge rates. The matrix S converts edge rates into node rates. Every term therefore has unit X/T .

The edge controller remains

$$J_e = M_e[f_e - \theta_e]_+, \quad f_e = \mu_i - \eta_e \mu_j. \quad (38.3)$$

All quantities are evaluated at the same instantaneous state $x(t)$.

38.2 The chain rule for the field

Because V depends on time through the changing state $x(t)$, the multivariable chain rule gives

$$\dot{V} = \nabla V(x)^\top \dot{x} = \mu(x)^\top \dot{x}. \quad (38.4)$$

Written coordinate by coordinate,

$$\dot{V} = \sum_{i=1}^m \frac{\partial V}{\partial x_i} \dot{x}_i = \sum_{i=1}^m \mu_i \dot{x}_i. \quad (38.5)$$

This form makes the units transparent. μ_i has unit B/X and $\dot{x}_i$ has unit X/T , so each product and the sum have unit B/T .

Substitute Equation (38.2):

$$\dot{V} = \mu^\top (u + SJ) \quad (38.6)$$

$$= \mu^\top u + \mu^\top SJ. \quad (38.7)$$

The first term is the contribution of natural or external drive. The second is the contribution of edge flow.

38.3 Turning the matrix term into edge forces

Matrix multiplication can be associated in either order:

$$\mu^\top SJ = (\mu^\top S)J. \quad (38.8)$$

Column e of S is S_e . From Equation (29.12),

$$\mu^\top S_e = -f_e. \quad (38.9)$$

Therefore the row vector $\mu^\top S$ is

$$\mu^\top S = (-f_1, -f_2, \dots, -f_{|E|}) = -f^\top. \quad (38.10)$$

Multiplying by J gives

$$\mu^\top SJ = -f^\top J = -\sum_e f_e J_e. \quad (38.11)$$

The product $f_e J_e$ has unit

$$\frac{B}{X} \frac{X}{T} = \frac{B}{T}. \quad (38.12)$$

It is a burden-reduction rate along edge e , analogous in mathematical form to force times flux in irreversible thermodynamics.

38.4 Using the thresholded mobility law

There are two cases for each edge.

If $f_e \leq \theta_e$, then $[f_e - \theta_e]_+ = 0$ and $J_e = 0$. Consequently $f_e J_e = 0$.

If $f_e > \theta_e$, then

$$J_e = M_e(f_e - \theta_e). \quad (38.13)$$

Solve this equation for f_e :

$$f_e = \frac{J_e}{M_e} + \theta_e. \quad (38.14)$$

Multiply by J_e :

$$f_e J_e = \frac{J_e^2}{M_e} + \theta_e J_e. \quad (38.15)$$

Both terms are non-negative because $M_e > 0$, $\theta_e \geq 0$, and $J_e \geq 0$. The same equation also holds on an inactive edge because all terms containing J_e are zero.

38.5 The continuous identity

Insert Equations (38.11) and (38.15) into Equation (38.7):

$$\dot{V} = \mu^\top u - \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right). \quad (38.16)$$

This is an identity for the declared dynamics, not an approximation. It separates two effects:

$\mu^\top u$ the field-change rate caused by natural or external drive;

$\sum_e (J_e^2/M_e + \theta_e J_e)$ the non-negative dissipation rate of the derived edge flow.

38.6 The undriven theorem

If $u = 0$, the identity becomes

$$\dot{V} = - \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right) \leq 0. \quad (38.17)$$

Thus V is non-increasing along every trajectory of the derived undriven continuous law for which the calculation is well-defined.

The phrase *non-increasing* is exact. It allows $\dot{V} = 0$. Equality holds when every generated flux is zero. That can happen at many states because viable bands are flat and thresholds can deactivate edges. The theorem does not say that the state converges to one unique point.

38.7 Numerical two-cell example

Let the current marginals be

$$\mu_i = -1, \quad \mu_j = -5, \quad \eta = 0.8. \quad (38.18)$$

Then

$$f = (-1) - 0.8(-5) = 3. \quad (38.19)$$

Choose $M = 2$ and $\theta = 1$. The flux is

$$J = 2[3 - 1]_+ = 4. \quad (38.20)$$

The edge dissipation rate can be calculated in two equivalent ways:

$$fJ = 3 \times 4 = 12, \quad (38.21)$$

$$\frac{J^2}{M} + \theta J = \frac{16}{2} + 1 \times 4 = 8 + 4 = 12. \quad (38.22)$$

With no drive, $\dot{V} = -12$ burden-units per unit time at that instant. This does not mean V will become negative. As the state changes, the marginals, force, and flux change; flow stops when the local force no longer exceeds the threshold.

38.8 A driven example

Keep the same edge dissipation 12. Suppose external demand and regeneration together give $\mu^\top u = 7$. Then

$$\dot{V} = 7 - 12 = -5. \quad (38.23)$$

The edge flow more than compensates for the drive at that instant.

If instead $\mu^\top u = 15$, then

$$\dot{V} = 15 - 12 = 3 > 0. \quad (38.24)$$

The controller still dissipates 12 units per time, but the external drive creates burden faster. The theorem has not failed. The driven identity reports the competition honestly.

38.9 Counterexample: arbitrary circulation

It is false that every local flow decreases V . Consider a lossless directed cycle with an externally imposed circulation $J \neq 0$ such that $SJ = 0$. The

state does not change, so $\dot{V} = 0$, even though edge rates are nonzero. If an arbitrary imposed flow moves against the gradient, it can increase V .

Why does this not contradict Equation (38.17)? Because that equation used the thresholded mobility relation between J and f . Arbitrary circulation does not necessarily satisfy it. The theorem concerns a specific gradient-derived controller, not all possible local actions.

38.10 Potential descent and useful service

A decrease in V is a mathematical property of the represented field. It does not guarantee that all demand was served. If physical resources are insufficient, the lowest-burden feasible trajectory can still contain unmet need. Conversely, an emergency action may temporarily increase V while serving a critical need.

The model must therefore record both field burden and functional outcomes. Treating one scalar as a complete social objective would erase the distinction the project is designed to preserve.

38.11 Connection to Lyapunov reasoning

A Lyapunov function is a scalar function used to study stability. A non-negative V with $\dot{V} \leq 0$ is a candidate Lyapunov function for the undriven derived dynamics. To infer convergence to a target set, additional facts are usually needed: bounded trajectories, forward invariance, regularity, and a characterization of the largest invariant subset where $\dot{V} = 0$.

The continuous identity supplies a powerful ingredient. It is not, by itself, a complete infinite-horizon homeostasis theorem for driven, constrained, multi-actor worlds.

Scope statement

Continuous dissipation is proved for the declared differentiable potential, loss-aware edge matrix, and thresholded mobility law, with all quantities evaluated at the same instantaneous state. Undriven monotonicity does not cover arbitrary flows, external actions, discontinuous settlement rules, or unrepresented fields.

Chapter recap

The chain rule converts state dynamics into field dynamics. The transpose of the edge matrix turns marginals into negative edge forces. The mobility law turns force times flux into a sum of non-negative dissipation terms. In an undriven system the declared potential cannot increase; in a driven system the drive and dissipation compete.

Exercises

1. Starting from $\dot{V} = \mu^\top \dot{x}$, reproduce every line leading to Equation (38.16).
2. Check the units of J^2/M using the units of mobility given earlier.
3. With $f = 2.5$, $M = 3$, and $\theta = 1$, calculate J , fJ , and $J^2/M + \theta J$.
4. Give one example in which $\dot{V} > 0$ in a driven system even though the edge flow follows the theorem.
5. Explain why $\dot{V} = 0$ does not imply one unique equilibrium state.

Discrete time, curvature, and the safe timestep

Computer models and transactional systems update in finite steps. A continuous law that decreases V can overshoot when implemented with a timestep that is too large. The discrete theorem quantifies this risk. It keeps the favourable first-order dissipation term and adds a second-order allowance for curvature.

39.1 Forward Euler as a finite update

For tick n , freeze the state x^n , calculate $u^n = u(x^n)$ and $J^n = J(x^n)$, and define

$$y^n = u^n + SJ^n. \quad (39.1)$$

The synchronous forward-Euler update is

$$x^{n+1} = x^n + \Delta t y^n. \quad (39.2)$$

y^n is a rate vector with unit X/T . The finite change $\Delta x^n = \Delta t y^n$ has unit X .

Forward Euler follows the tangent direction at the beginning of the tick. If the tick is long, the true field slope can change materially before the finite step ends. Curvature controls the difference between the tangent prediction and the exact field value at the new state.

39.2 What a Lipschitz gradient means

Assume the gradient satisfies

$$\|\nabla V(x) - \nabla V(y)\|_2 \leq L_V \|x - y\|_2 \quad (39.3)$$

through the relevant region. The constant L_V has unit B/X^2 . It bounds how quickly the marginal vector can change as the state changes.

For the separable squared-hinge potential, a safe global bound within one homogeneous field is

$$L_V = \max_i 2(\alpha_i + \beta_i + \chi_i), \quad (39.4)$$

although a sharper bound may use only penalty branches active on the represented region. A valid upper bound may be conservative; an underestimated bound makes the certificate unsafe.

39.3 The descent lemma

For a function with L_V -Lipschitz gradient, the descent lemma states

$$V(x + \Delta x) \leq V(x) + \nabla V(x)^\top \Delta x + \frac{L_V}{2} \|\Delta x\|_2^2. \quad (39.5)$$

The first added term is the tangent prediction. The final term is a guaranteed curvature allowance. It is quadratic in the size of the step.

Substitute $\Delta x = \Delta t y$:

$$V(x^{n+1}) - V(x^n) \leq \Delta t \mu^\top y + \frac{L_V \Delta t^2}{2} \|y\|_2^2. \quad (39.6)$$

The factor Δt^2 is why decreasing the timestep eventually makes the first-order term dominate whenever there is strict instantaneous dissipation.

39.4 Insert drive and edge dissipation

Since $y = u + SJ$,

$$\mu^\top y = \mu^\top u + \mu^\top SJ \quad (39.7)$$

$$= \mu^\top u - \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right). \quad (39.8)$$

Insert this into Equation (39.6):

$$V(x^{n+1}) - V(x^n) \leq \Delta t \mu^\top u \quad (39.9)$$

$$- \Delta t \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right) \quad (39.10)$$

$$+ \frac{L_V \Delta t^2}{2} \|u + SJ\|_2^2. \quad (39.11)$$

The three lines mean:

1. external or natural drive can raise or lower burden;
2. the generated edge flow contributes first-order dissipation;
3. finite stepping adds a non-negative curvature allowance.

39.5 The undriven one-edge certificate

Consider one active edge, $u = 0$, and $J > 0$. The update direction is $S_e J$. Its squared norm is

$$\|S_e J\|_2^2 = J^2 \|S_e\|_2^2 = J^2 (1 + \eta_e^2). \quad (39.12)$$

The last equality follows because S_e has entries -1 and η_e on two different coordinates:

$$\|S_e\|_2^2 = (-1)^2 + \eta_e^2 = 1 + \eta_e^2. \quad (39.13)$$

Ignoring the additional favourable threshold term gives the sufficient bound

$$V(x^{n+1}) - V(x^n) \leq -\Delta t \frac{J^2}{M_e} + \frac{L_V \Delta t^2}{2} J^2 (1 + \eta_e^2). \quad (39.14)$$

Factor out the non-negative $\Delta t J^2$:

$$V(x^{n+1}) - V(x^n) \leq \Delta t J^2 \left[-\frac{1}{M_e} + \frac{L_V \Delta t}{2} (1 + \eta_e^2) \right]. \quad (39.15)$$

The bracket is non-positive when

$$\Delta t \leq \frac{2}{L_V M_e (1 + \eta_e^2)}. \quad (39.16)$$

Check the unit. $L_V M_e$ has unit

$$\frac{B}{X^2} \frac{X^2}{BT} = \frac{1}{T}, \quad (39.17)$$

so the right side has unit T , as a timestep must.

39.6 A numerical certificate

Let $L_V = 4$, $M = 0.5$, and $\eta = 0.8$. Then

$$\Delta t_{\max} = \frac{2}{4 \times 0.5 \times (1 + 0.8^2)} \quad (39.18)$$

$$= \frac{2}{2 \times 1.64} \quad (39.19)$$

$$\approx 0.6098. \quad (39.20)$$

A registered timestep of 0.3 lies below the sufficient certificate. A timestep of 0.8 lies above it. Being above the bound does not prove that this particular tick will fail; it means the theorem no longer guarantees descent. The distinction between sufficient and necessary conditions is important.

39.7 A pure quadratic overshoot

Take the simple potential

$$V(x) = \frac{1}{2}x^2 \quad (39.21)$$

and continuous descent law $\dot{x} = -kx$ with $k > 0$. Then

$$\dot{V} = x\dot{x} = -kx^2 \leq 0. \quad (39.22)$$

Forward Euler gives

$$x^{n+1} = (1 - k\Delta t)x^n. \quad (39.23)$$

Therefore

$$V(x^{n+1}) = \frac{1}{2}(1 - k\Delta t)^2(x^n)^2. \quad (39.24)$$

The potential does not increase exactly when

$$|1 - k\Delta t| \leq 1, \quad (39.25)$$

which is equivalent to $0 \leq k\Delta t \leq 2$. For a larger step, the discrete state jumps across the minimum with increased magnitude. The continuous law is stable, but its coarse discretisation is not.

This elementary example explains why a timestep certificate is a scientific part of the discrete model rather than a mere programming preference.

39.8 Exact remainder and upper bound

The descent lemma gives an upper bound on the finite remainder. In a test we can also calculate the exact remainder

$$R_{\text{exact}} = V(x + \Delta x) - V(x) - \mu(x)^\top \Delta x. \quad (39.26)$$

The theorem requires

$$R_{\text{exact}} \leq \frac{L_V}{2} \|\Delta x\|_2^2. \quad (39.27)$$

These two numbers should not be confused. Nor is either one the stock-ledger residual. The exact remainder measures nonlinearity of V over one state step; the ledger residual measures whether physical inputs, outputs, and losses add up.

39.9 The driven sufficient condition

In a driven system, Equation (39.11) guarantees non-increase if

$$\sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right) \geq \mu^\top u + \frac{L_V \Delta t}{2} \|u + SJ\|_2^2. \quad (39.28)$$

The edge dissipation must cover both the first-order drive contribution and the finite-step curvature allowance. If it does not, V may increase even though the code correctly implements the derived edge law.

39.10 What the theorem excludes

The proof assumes a frozen synchronous update. State projection, clipping, sequential edge updates, mid-tick reclassification, or replacing accepted flow with a different executed flow changes the mathematical map. Those operations may be useful, but their effects need separate analysis.

A projected non-negative state can produce a visually plausible trajectory while hiding requested service that was not physically available. Closing a post-hoc stock ledger is not enough; the service and acceptance ledgers must also distinguish request, feasible acceptance, delivery, unmet need, and loss.

Scope statement

The discrete inequality is a sufficient theorem for the declared synchronous Euler map and a valid curvature bound. A certificate below which descent is guaranteed need not be the largest stable timestep. A run above the certificate is outside the guarantee, not automatically a counterexample. A run below it still says nothing by itself about service or infinite-horizon stability.

Chapter recap

Finite stepping adds a curvature term of order Δt^2 . The one-edge certificate makes the favourable first-order dissipation dominate this remainder. Timestep bounds have physical units and mathematical meaning. Sufficient bounds certify a region; crossing them removes the guarantee but does not prove immediate failure.

Exercises

1. Verify the units of every term in Equation (39.11).
2. Recalculate the numerical one-edge certificate for $L_V = 3$, $M = 2$, and $\eta = 0.5$.
3. For the quadratic overshoot example with $k = 4$, compare the updates at $\Delta t = 0.2$, 0.5 , and 0.75 .
4. Explain the difference between an exact curvature remainder, a curvature upper bound, and a stock-ledger residual.
5. Why can projection change the theorem even if it prevents negative numerical states?

Network geometry and multi-edge timestep certificates

The one-edge calculation shows where efficiency and mobility enter a timestep bound. A network introduces overlap: several edges may share a source or destination, so their state-change vectors can reinforce or partially cancel. The matrix norm in the network certificate measures this geometry.

40.1 Mobility as a diagonal matrix

Collect edge mobilities into

$$D_M = \text{diag}(M_1, \dots, M_{|E|}). \quad (40.1)$$

The word “diag” means a square matrix whose listed values lie on the diagonal and whose off-diagonal entries are zero. Its square root is

$$D_M^{1/2} = \text{diag}(\sqrt{M_1}, \dots, \sqrt{M_{|E|}}). \quad (40.2)$$

Similarly,

$$D_M^{-1/2} = \text{diag}(1/\sqrt{M_1}, \dots, 1/\sqrt{M_{|E|}}). \quad (40.3)$$

These matrices let us compare the state-change size $\|SJ\|$ with the mobility-weighted dissipation $\sum_e J_e^2/M_e$.

40.2 The spectral norm in plain language

For a matrix A , the spectral norm is

$$\|A\|_2 = \max_{y \neq 0} \frac{\|Ay\|_2}{\|y\|_2}. \quad (40.4)$$

It is the largest factor by which the matrix can stretch a vector in Euclidean length. It does not mean that the system literally chooses the worst vector on every tick. It supplies a conservative bound valid for every possible vector.

Set

$$A = SD_M^{1/2}, \quad y = D_M^{-1/2}J. \quad (40.5)$$

Then

$$Ay = SD_M^{1/2}D_M^{-1/2}J = SJ. \quad (40.6)$$

By the definition of the spectral norm,

$$\|SJ\|_2^2 \leq \|SD_M^{1/2}\|_2^2 \|D_M^{-1/2}J\|_2^2. \quad (40.7)$$

The final norm expands to

$$\|D_M^{-1/2}J\|_2^2 = \sum_e \frac{J_e^2}{M_e}. \quad (40.8)$$

Thus the same quantity that appears in dissipation also controls the maximum network state-change norm.

40.3 The global spectral certificate

In the undriven discrete inequality, discard the favourable threshold terms and apply Equation (40.7):

$$V(x^{n+1}) - V(x^n) \leq -\Delta t \sum_e \frac{J_e^2}{M_e} \quad (40.9)$$

$$+ \frac{L_V \Delta t^2}{2} \|SD_M^{1/2}\|_2^2 \sum_e \frac{J_e^2}{M_e}. \quad (40.10)$$

Factor the weighted flux sum:

$$V(x^{n+1}) - V(x^n) \leq \Delta t \left(\sum_e \frac{J_e^2}{M_e} \right) \left[-1 + \frac{L_V \Delta t}{2} \|SD_M^{1/2}\|_2^2 \right]. \quad (40.11)$$

A sufficient network-wide certificate is therefore

$$\Delta t \leq \frac{2}{L_V \|SD_M^{1/2}\|_2^2}. \quad (40.12)$$

For a graph containing only one edge, the squared norm becomes $M_e(1 + \eta_e^2)$ and the result reduces to the one-edge certificate.

40.4 Why shared endpoints tighten the bound

Suppose one source has four outgoing edges. Each edge column contains -1 in the same source row. When all four rates are positive, their source withdrawals add. The matrix can therefore produce a large state-change norm from a given mobility-weighted rate vector. The spectral norm detects this overlap.

If the edges acted on disjoint pairs of nodes, their columns would be more orthogonal and the worst amplification could be smaller. Network topology matters not because a global controller reads every state, but because local edge effects can share coordinates.

40.5 Active-set certificates

The full matrix includes every permitted edge, even when many are inactive at the current state. If only a subset A_E has $J_e > 0$, form the restricted matrix S_{A_E} and restricted mobility matrix D_{M,A_E} . Then

$$\Delta t \leq \frac{2}{L_V \|S_{A_E} D_{M,A_E}^{1/2}\|_2^2} \quad (40.13)$$

can be less conservative.

The active set must be determined from the frozen state used for the update. Recomputing it halfway through a synchronous tick would change the map being certified.

40.6 A state-specific certificate

For a particular nonzero flux vector in the undriven case, define the exact instantaneous dissipation

$$D(J) = \sum_e \left(\frac{J_e^2}{M_e} + \theta_e J_e \right). \quad (40.14)$$

The discrete inequality gives descent whenever

$$-\Delta t D(J) + \frac{L_V \Delta t^2}{2} \|SJ\|_2^2 \leq 0. \quad (40.15)$$

For $SJ \neq 0$, divide by Δt and solve:

$$\Delta t \leq \frac{2D(J)}{L_V \|SJ\|_2^2}. \quad (40.16)$$

For a nonzero flux produced by the present controller, SJ cannot equal zero. To see why, suppose $SJ = 0$. Then

$$0 = \mu^\top SJ = -D(J). \quad (40.17)$$

Since every term in $D(J)$ is non-negative, $D(J) = 0$ forces every $J_e = 0$, contradicting the assumption of nonzero generated flux.

40.7 Gershgorin and degree-based alternatives

Computing a spectral norm is a global mathematical operation on the declared graph. It can be performed during model certification even when runtime decisions remain local. When a transparent conservative bound is preferred, row sums or Gershgorin-disc arguments can bound the largest eigenvalue of the relevant weighted matrix using node degrees and edge weights.

Such a bound is usually easier to audit but more conservative. The important scientific requirement is not that one method always be used; it is that the chosen bound be valid, reproducible, and locked before the result it is meant to certify.

40.8 A small fan-out calculation

Let one source feed two destinations with efficiencies 1 and equal mobilities $M = 1$. The edge matrix is

$$S = \begin{pmatrix} -1 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (40.18)$$

Then

$$S^\top S = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}. \quad (40.19)$$

The eigenvalues are 3 and 1, so $\|S\|_2^2 = 3$. If $L_V = 2$, the network certificate is

$$\Delta t \leq \frac{2}{2 \times 3} = \frac{1}{3}. \quad (40.20)$$

Each edge alone would give $\Delta t \leq 1/2$. The shared source tightens the multi-edge guarantee from 0.5 to approximately 0.333.

40.9 Local decision versus global certification

The presence of a spectral certificate does not mean each actor reads the whole global state. The graph and parameter set can be certified at design time. Runtime action selection can still read only permitted local endpoints, the local source budget, and declared edge parameters.

This distinction appears throughout the project: global mathematics can prove a property of a local rule without becoming a global decision input. A local thermostat can be analysed using a global building model; the thermostat does not need to solve that model every second.

Scope statement

The spectral, active-set, degree-based, and state-specific certificates are sufficient bounds for the declared synchronous update. They rely on compatible coordinate scaling and a valid curvature bound. They do not establish physical feasibility, reserve preservation, service success, or optimality of the chosen graph.

Chapter recap

The matrix norm measures how overlapping edge columns can amplify a flux vector into node-state change. Mobility weighting links this geometry to dissipation. The spectral certificate is global in proof but need not be global in runtime information. Active-set and state-specific versions can be sharper; transparent degree bounds can be more conservative.

Exercises

1. Verify the eigenvalues of the two-edge fan-out matrix $S^\top S$.
2. Explain in words why two edges sharing a source are not geometrically independent.
3. Derive Equation (40.16) from the undriven discrete inequality.
4. Why can a design-time global certificate coexist with a local runtime decision rule?
5. Give one reason an active-set certificate can be larger than a full-graph certificate.

One-step preservation, feasibility, and genuine need

Potential descent and source preservation are related but not identical. A transfer can lower the total represented burden while withdrawing too much from one regenerative source in one coarse tick. Conversely, a preservation gate can keep a source above its reserve while leaving demand unmet. This chapter derives the local export budget that separates those questions.

41.1 The source-side update

Consider a protected source node i . Let

x_i be its frozen starting stock;

u_i be its frozen natural or exogenous rate before actor export;

R_i^{eff} be the effective protected reserve, including any declared safety offset;

Q_i be the aggregate accepted outgoing rate during the tick;

I_i be the aggregate usable incoming rate during the tick.

Its synchronous update is

$$x_i^{n+1} = x_i + \Delta t(u_i + I_i - Q_i). \quad (41.1)$$

Incoming flow satisfies $I_i \geq 0$ in the simple stock model. For a conservative export permission, ignore that incoming help when calculating how much the source may release. Require

$$x_i + \Delta t(u_i - Q_i) \geq R_i^{\text{eff}}. \quad (41.2)$$

This is a one-step boundary condition, not a potential argument.

41.2 Deriving the safe export budget

Begin with Equation (41.2):

$$x_i + \Delta t u_i - \Delta t Q_i \geq R_i^{\text{eff}}. \quad (41.3)$$

Move the export term to the right and the reserve to the left:

$$x_i + \Delta t u_i - R_i^{\text{eff}} \geq \Delta t Q_i. \quad (41.4)$$

Because $\Delta t > 0$, divide without reversing the inequality:

$$Q_i \leq \frac{x_i + \Delta t u_i - R_i^{\text{eff}}}{\Delta t}. \quad (41.5)$$

An export rate cannot be negative. Define

$$Q_{i,\max} = \frac{[x_i + \Delta t u_i - R_i^{\text{eff}}]_+}{\Delta t}. \quad (41.6)$$

The numerator is the amount by which the no-export successor lies above the effective reserve. Dividing by tick duration converts that finite stock margin into an allowable rate.

41.3 Why the positive part needs a feasibility flag

Suppose

$$x_i + \Delta t u_i < R_i^{\text{eff}}. \quad (41.7)$$

The source crosses the reserve even with $Q_i = 0$. The positive part in Equation (41.6) returns zero, correctly permitting no export, but zero export does not make the next state safe. The state is locally infeasible under the frozen no-export dynamics.

A robust implementation must therefore report both

1. the permitted export rate, which is zero; and
2. the no-export infeasibility, which remains a material domain outcome.

Silently treating every zero budget as successful preservation would hide natural collapse or excessive imposed demand.

41.4 A four-state classification

One useful source-side classification is:

Protected and feasible (P) current stock is at or above the effective reserve and the no-export successor remains at or above it. A regenerative source may use the robust aggregate export budget. A finite or irreversible source receives zero preservation-safe export; extraction would be depletion, not preservation.

Recovery (R) current stock is below the reserve. Protected export is zero while recovery is required.

Locally infeasible (I) current stock is not yet below the reserve, but the no-export successor crosses it. Export is zero and the failure is reported.

Declared flow source (F) the represented source is a certified flow rather than a protected stock. Its budget is the smaller of its declared flow cap and the non-negative available-stock rate, $\min\{\text{flow cap}, [x_i + \Delta tu_i]_+ / \Delta t\}$.

The labels are an implementation aid. The theorem depends on the inequalities, not on the letters.

41.5 A numerical feasible case

Let

$$x_i = 10, \quad R_i^{\text{eff}} = 8, \quad u_i = -0.5 \text{ units/hour}, \quad \Delta t = 2 \text{ hours}. \quad (41.8)$$

Without export, the successor is

$$x_i + \Delta tu_i = 10 + 2(-0.5) = 9, \quad (41.9)$$

which is one unit above reserve. The maximum export rate is

$$Q_{i,\max} = \frac{9-8}{2} = 0.5 \text{ units/hour.} \quad (41.10)$$

At the full budget, the finite exported quantity is one unit and the conservative next stock is exactly eight.

If an incoming delivery of 0.2 units/hour also occurs, the actual next stock is

$$10 + 2(-0.5 + 0.2 - 0.5) = 8.4. \quad (41.11)$$

The incoming flow creates extra margin because it was not relied on when the export budget was certified.

41.6 A numerical infeasible case

Keep $x_i = 10$, $R_i^{\text{eff}} = 8$, and $\Delta t = 2$, but let $u_i = -1.2$ units/hour. The no-export successor is

$$10 + 2(-1.2) = 7.6 < 8. \quad (41.12)$$

Equation (41.6) returns zero. Nevertheless the expected successor is below the boundary. The correct report is not “the gate preserved the source”. It is “no export was allowed, and the source was locally infeasible under the declared drive”.

This distinction matters in a hospital system. If essential local consumption alone exceeds replenishment, refusing external shipments does not solve the shortage. The model should expose the need for emergency supply, demand management, substitution, or structural change.

41.7 Measurement margins

Suppose measured stock may overstate true stock by at most ε_x and the estimated drive may overstate true drive by at most ε_u . Use the conservative values $x_i - \varepsilon_x$ and $u_i - \varepsilon_u$:

$$Q_{i,\max}^{\text{rob}} = \frac{[(x_i - \varepsilon_x) + \Delta t(u_i - \varepsilon_u) - R_i^{\text{eff}}]_+}{\Delta t}. \quad (41.13)$$

This is not the same as a floating-point tolerance. ε_x and ε_u are physical uncertainty margins with units. A numerical tolerance only decides whether two finite representations should be treated as equal for diagnostics. It must not create material export capacity.

For a robust preservation certificate, the conservative current stock and conservative no-export successor must themselves be feasible:

$$x_i - \varepsilon_x \geq R_i^{\text{eff}}, \quad (x_i - \varepsilon_x) + \Delta t(u_i - \varepsilon_u) \geq R_i^{\text{eff}}. \quad (41.14)$$

If either inequality fails, the positive part may return a zero export budget, but refusing export does not prove preservation under the declared uncertainty envelope. Nominal State P and robust theorem eligibility are therefore distinct when the physical margins are nonzero.

41.8 Several outgoing requests

Let $\mathcal{O}(i)$ be the set of edges leaving source i . Each edge requests q_e^{req} . The aggregate request is

$$Q_i^{\text{req}} = \sum_{e \in \mathcal{O}(i)} q_e^{\text{req}}. \quad (41.15)$$

Write $Q_{i,\max}$ for the applicable budget returned after classification. It is $Q_{i,\max}^{\text{rob}}$ for a regenerative class-P source, the flow-cap/available-stock minimum for class F, and zero for class R, class I, or a finite or irreversible class-P source.

The released controller uses the following fail-closed diagnostic and scaling convention:

$$\sigma_i = \begin{cases} \min \left(1, \frac{Q_{i,\max}}{Q_i^{\text{req}}} \right), & Q_i^{\text{req}} > 0 \text{ and } Q_{i,\max} > 0, \\ 0, & \text{otherwise.} \end{cases} \quad (41.16)$$

Thus a zero aggregate request is accepted as zero and recorded with $\sigma_i = 0$; a positive request against a zero budget is also scaled to zero. The value of σ_i in the zero-request branch cannot change a physical quantity, but matching the released convention makes diagnostic traces unambiguous.

Every outgoing request is scaled from the same frozen factor:

$$q_e^{\text{acc}} = \sigma_i q_e^{\text{req}}. \quad (41.17)$$

Then

$$\sum_{e \in \mathcal{O}(i)} q_e^{\text{acc}} = \sigma_i Q_i^{\text{req}} \leq Q_{i,\max}. \quad (41.18)$$

Proportional scaling is a physically order-independent way to respect the aggregate budget. It is not asserted to be the best social priority rule.

41.9 The one-step preservation theorem

Theorem: aggregate reserve preservation. Assume

1. $x_i \geq R_i^{\text{eff}}$;
2. the no-export successor satisfies $x_i + \Delta t u_i \geq R_i^{\text{eff}}$;
3. the applicable protected-source budget obeys $0 \leq Q_{i,\max} \leq [x_i + \Delta t u_i - R_i^{\text{eff}}]/\Delta t$;
4. accepted aggregate export obeys $Q_i^{\text{acc}} \leq Q_{i,\max}$;
5. incoming usable flow is non-negative;
6. the synchronous update uses the same frozen quantities.

Then $x_i^{n+1} \geq R_i^{\text{eff}}$.

Proof. From Equation (41.1),

$$x_i^{n+1} = x_i + \Delta t(u_i + I_i - Q_i^{\text{acc}}) \quad (41.19)$$

$$\geq x_i + \Delta t(u_i - Q_i^{\text{acc}}) \quad \text{because } I_i \geq 0, \quad (41.20)$$

$$\geq x_i + \Delta t(u_i - Q_{i,\max}) \quad \text{because } Q_i^{\text{acc}} \leq Q_{i,\max}, \quad (41.21)$$

$$\geq R_i^{\text{eff}}. \quad (41.22)$$

The final inequality follows from the no-export feasibility and budget assumptions. $\square$

For the robust version, additionally assume Equation (41.14), $x_i^{\text{true}} \geq x_i - \varepsilon_x$, $u_i^{\text{true}} \geq u_i - \varepsilon_u$, and $Q_i^{\text{acc}} \leq Q_{i,\max}^{\text{rob}}$. Then

$$x_i^{n+1} \geq (x_i - \varepsilon_x) + \Delta t(u_i - \varepsilon_u - Q_{i,\max}^{\text{rob}}) \geq R_i^{\text{eff}}. \quad (41.23)$$

If conservative no-export feasibility fails, $Q_{i,\max}^{\text{rob}} = 0$ is a refusal result, not a robust preservation conclusion.

41.10 Why the theorem is an invariant implication

The theorem has the logical form

$$x^n \in \mathcal{K} \text{ and assumptions hold at tick } n \implies x^{n+1} \in \mathcal{K}, \quad (41.24)$$

where $\mathcal{K}$ contains states above the protected reserve. If the same conditions are re-established at every tick, induction can extend the boundary property over many ticks. But the no-export feasibility condition may cease to hold under a shock, changing demand, model error, or exhausted regeneration. The one-step theorem does not make that condition permanent.

41.11 Preservation is not service

Suppose one tick lasts $\Delta t = 1$ hour. Two hospitals each request an oxygen transport rate of 80 cylinders per hour, while the safe source budget is 100 cylinders per hour. Proportional scaling accepts 50 cylinders per hour on each edge. Let both edge efficiencies be $\eta_e = 0.9$. The transport ledger for this one-hour tick is therefore

$$\text{requested transport rate} = 160 \text{ cylinders/hour}, \quad (41.25)$$

$$\text{accepted transport rate} = 100 \text{ cylinders/hour}, \quad (41.26)$$

$$\text{sent amount} = \Delta t \sum_e q_e^{\text{acc}} = 100 \text{ cylinders}, \quad (41.27)$$

$$\text{transport-delivered amount} = \Delta t \sum_e \eta_e q_e^{\text{acc}} = 90 \text{ cylinders}, \quad (41.28)$$

$$\text{transport loss} = \Delta t \sum_e (1 - \eta_e) q_e^{\text{acc}} = 10 \text{ cylinders}. \quad (41.29)$$

Transport delivery is not yet the bounded-service quantity. Suppose each hospital already has $x_j = 5$ usable cylinders, has no other local drive during the tick ($u_j = 0$), and has demand rate $d_j = 80$ cylinders per hour. For each hospital the released bounded-service semantics give

$$\text{available}_j = x_j + \Delta t(u_j + \eta_j q_j^{\text{acc}}) = 5 + 1(0 + 0.9 \cdot 50) = 50, \quad (41.30)$$

$$\text{demand amount}_j = \Delta t d_j = 80, \quad (41.31)$$

$$\text{service}_j = \min(\text{demand amount}_j, \text{available}_j) = 50, \quad (41.32)$$

$$\text{unmet}_j = \text{demand amount}_j - \text{service}_j = 30. \quad (41.33)$$

Total service is 100 cylinders and total unmet demand is 60. Service exceeds the 90 cylinders delivered in transport because the hospitals began with ten cylinders of usable inventory. In a different state it could be capped below transport delivery by lower demand. This is why requested transport, accepted transport, delivered stock, available stock, service, and unmet need must remain distinct fields.

An honest model does not manufacture service by projecting negative stocks back to zero or by counting requested quantities as delivered quantities.

41.12 Need priority and emergency borrowing

The preservation theorem does not decide how scarce service is allocated. A medical priority rule may direct more of the accepted stock to the patient with the most urgent need. If serving that need requires a high-burden alternative source, the negative EBU should remain visible. A later EBU institution

could allow borrowing, insurance, mutual aid, or public assumption of the burden.

The account should never make an essential need appear physically cheap by deleting its cost. High EBU can be the signal that mobilises long-term redesign: more local production, more resilient inventory, lower-loss transport, or a different hospital network.

41.13 What descent does not add

Even if the global potential decreases during the tick, a source reserve can be violated if a larger improvement at the destination outweighs source damage in V . The direct export budget is therefore not redundant.

Conversely, preserving every source boundary does not guarantee that V decreases; external demand or excess at destinations may raise it. Preservation and descent are two compatible certificates aimed at different failure modes.

Scope statement

The theorem preserves one declared source reserve for one synchronous tick when no-export feasibility, measurement bounds, and aggregate accepted export conditions hold. It does not prove that demand is feasible, that the reserve certificate is ecologically correct, that proportional allocation is fair, or that the conditions hold forever.

Chapter recap

The safe export budget is the no-export stock margin above reserve divided by tick duration. A zero budget can mean either safe refusal or unavoidable local infeasibility, so the status must be reported. Multiple requests share one aggregate cap. Under explicit conditions the next source state remains above reserve, while service and priority remain separate questions.

Exercises

1. Derive Equation (41.6) from the reserve inequality without skipping algebraic steps.

2. For $x = 12$, $R^{\text{eff}} = 9$, $u = -0.4$, and $\Delta t = 2$, calculate the no-export successor and maximum export rate.
3. Repeat with $u = -1.8$. Classify the source and explain why zero export is not a successful preservation result.
4. Three requests are 4, 3, and 5 units/hour and the budget is 6. Calculate σ and all accepted rates.
5. Explain the difference between a physical uncertainty margin and a floating-point comparison tolerance.
6. Construct a case in which global V falls while one source reserve is crossed.